



We are on a mission to address the digital skills gap for 10 Million+ young professionals, train and empower them to forge a career path into future tech

COMPUTER NETWORKS

1-JULY-26



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- TCP/IP Model
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Fundamentals of Networking

Introduction

- A computer network is a **network of devices** like computers, servers, routers, switches, or other related hardware's that are **interlinked with one another** to communicate and share resources, exchange data, and shared services.



- Computer networks work by **transmitting data** between devices in the form of **small packets**. These packets travel across the network through **various mediums**, such as:
 - **Wired** connections (like Ethernet cables)
 - **Wireless** connections (like Wi-Fi, Bluetooth, or cellular networks)

Fundamentals of Networking

Examples

In Everyday Life:

- A **home network** that connects your laptop, smartphone, and smart TV.
- **Office** networks that allow employees to collaborate and share files.
- The **internet itself** — a massive network connecting billions of devices globally.

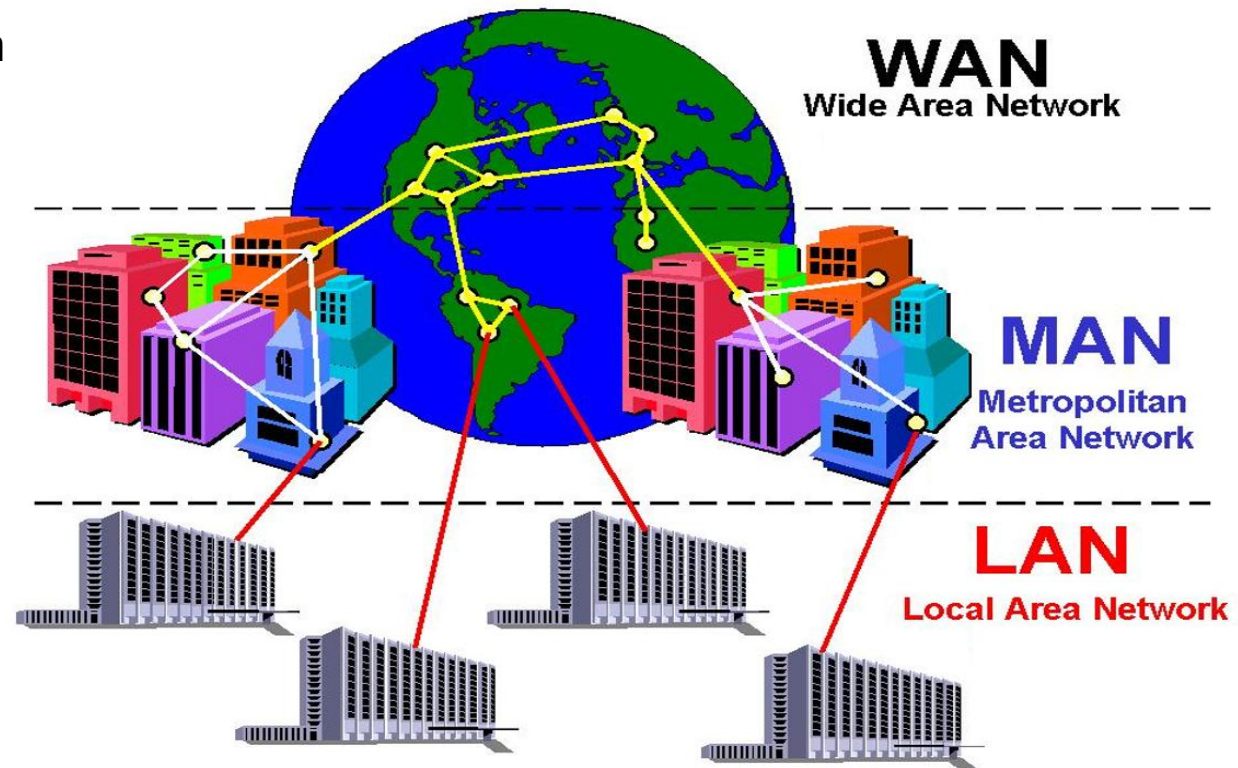
Computer networks have revolutionized how we communicate, work, and live, forming the foundation of modern information technology.

Benefits

- **Information sharing:** Authorized users can use other computers on the network to access and share information and data. This could include special group projects, databases, etc.
- **Hardware sharing:** One device connected to a network, such as a printer or scanner, can be shared by many users.
- **Software sharing:** Instead of purchasing and installing a software program on each computer, it can be installed on the server. All of the users can then access the program from a single location.
- **Collaborative environment:** Users can work together on group projects by combining the power and capabilities of diverse equipment.

Types of Computer Networks

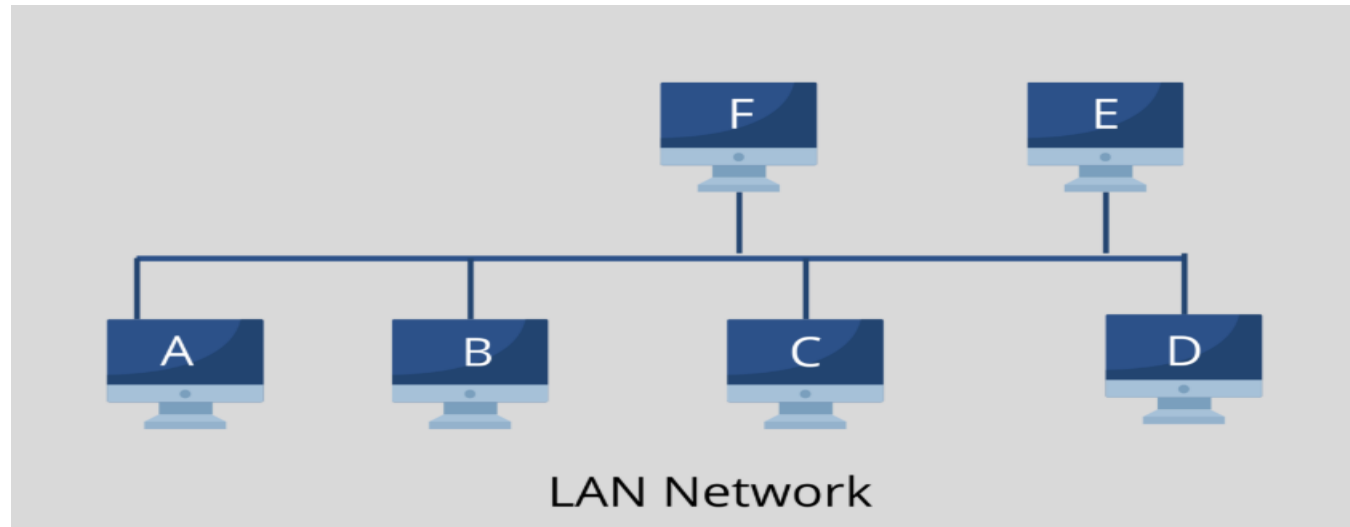
- Computer networks can be categorized based on various factors such as **geographical scope**, **architecture**, and the medium of **transmission**. Each type serves specific purposes, depending on the **scale and nature** of the operation



Types of Computer Networks

Local Area network(LAN)

- A Local Area Network (LAN) is a network that connects computers and devices **within a limited geographical area** such as a single building, office, or campus.
- LANs are typically used to share resources **like printers, files, and internet connections** among multiple devices.



Types of Computer Networks

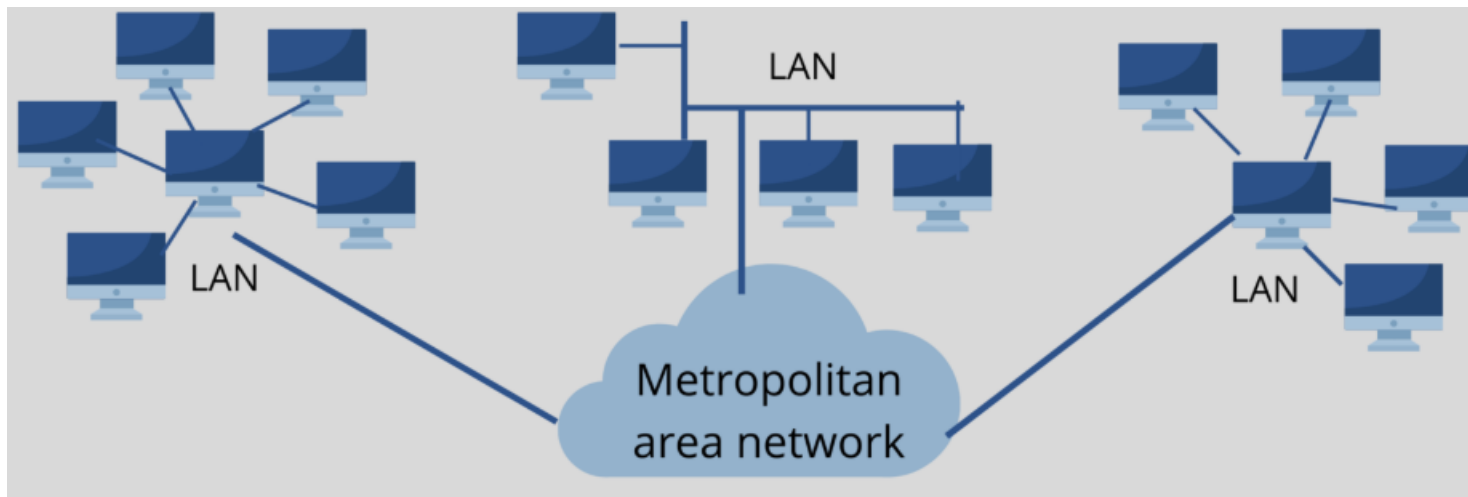
Features of LAN

- **High Data Transfer Speed:** LANs offer high data transfer rates, typically ranging from 100 Mbps to 10 Gbps.
- **Low Latency:** Due to the close proximity of connected devices, LANs exhibit low latency which means data travels quickly between nodes.
- **Limited Range:** They generally cover small geographic areas, making them ideal for localized networking needs.
- **Wired or Wireless Connectivity:** LANs can function using Ethernet cables (wired) or Wi-Fi (wireless) connections.
- **Centralized Administration:** Easier to administer and manage resources through local servers.

Types of Computer Networks

Metropolitan Area Network (MAN)

- A Metropolitan Area Network (MAN) spans a **larger geographical area than a LAN**, covering a city or a large campus. MANs are designed to extend and connect multiple LANs within a specific region.



Types of Computer Networks

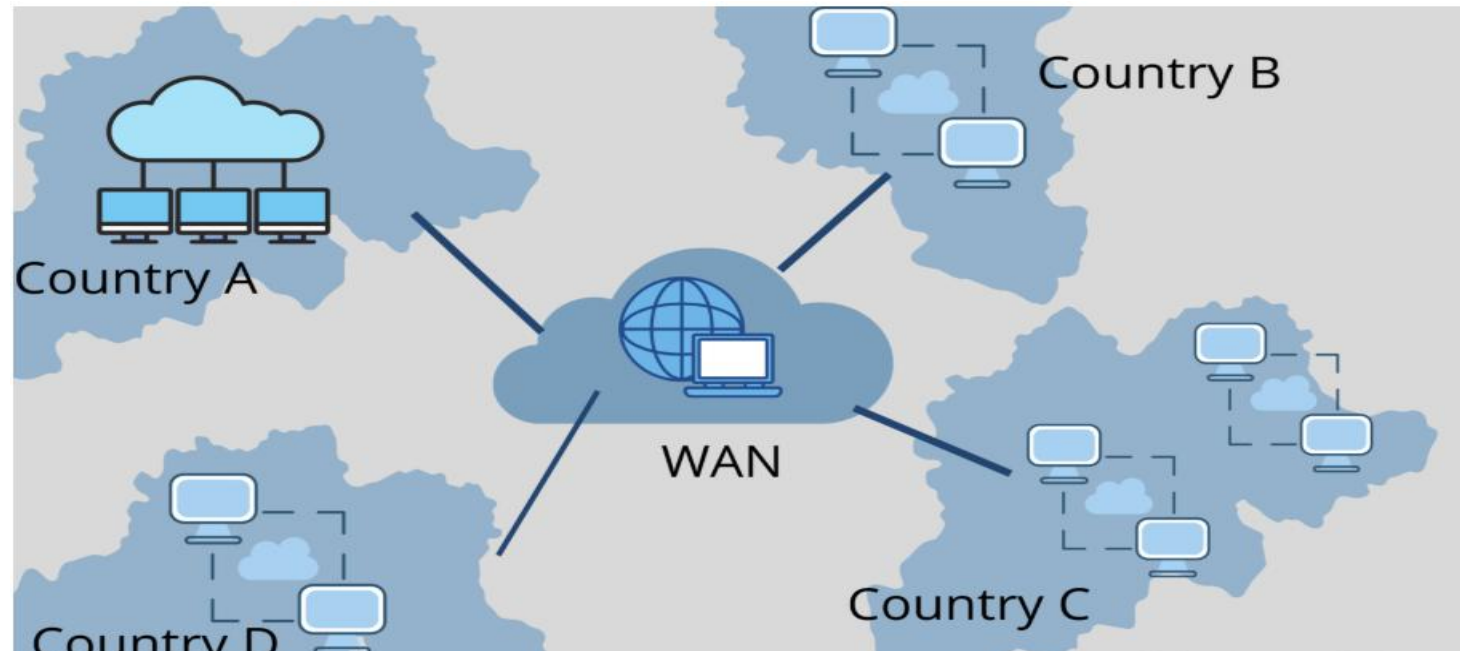
Features of MAN

- **Moderate Range:** Connects multiple LANs over distances up to 50 kilometers.
- **High-Capacity Backbone:** Utilizes high-capacity transmission media like fiber optics.
- **Enhanced Performance:** Provides higher data transfer speeds compared to WAN.
- **Integration:** Can integrate with other MANs and WANs for broader connectivity.

Types of Computer Networks

Wide Area Network (WAN)

- A Wide Area Network (WAN) covers a **broad geographic area, often a country or even globally.**
- WANs are designed to connect multiple LANs and MANs, enabling long-distance communication and resource sharing



Types of Computer Networks

Features of WAN

- **Extensive Coverage:** Can connect devices and networks across vast distances.
- **Diverse Connectivity:** Utilizes various transmission media, including satellites, leased lines, and public networks.
- **Complex Infrastructure:** Involves a more complex and hierarchical network structure.
- **Lower Data Transfer Speed:** Generally offers lower data transfer speeds compared to LANs and MANs due to longer distances and multiple traversed networks.

Fundamentals of Networking

Basic Components

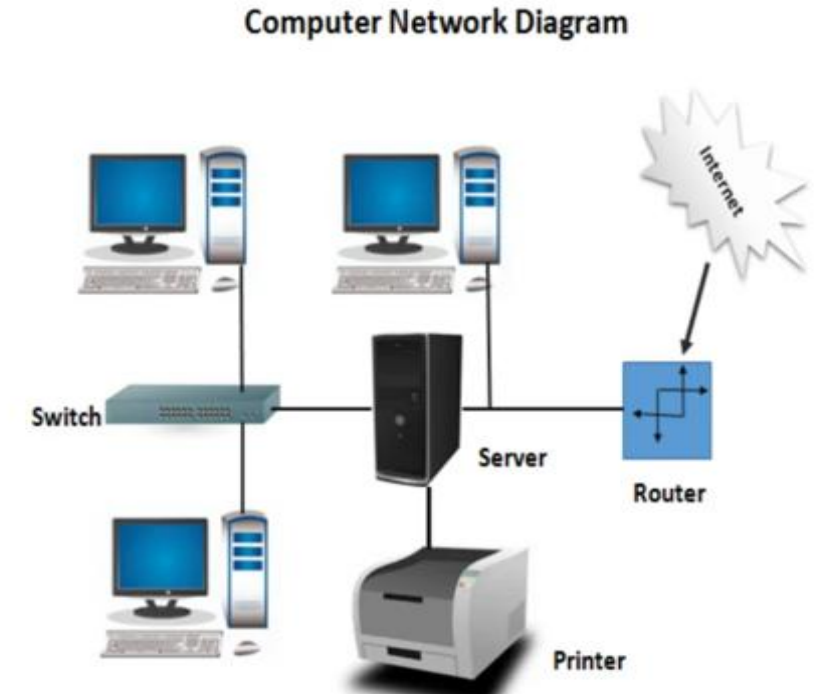
The basic components of a computer network can be classified into two types:

1) Hardware components :

- The physical equipment needed to **create a network** is known as the hardware components of a computer network.

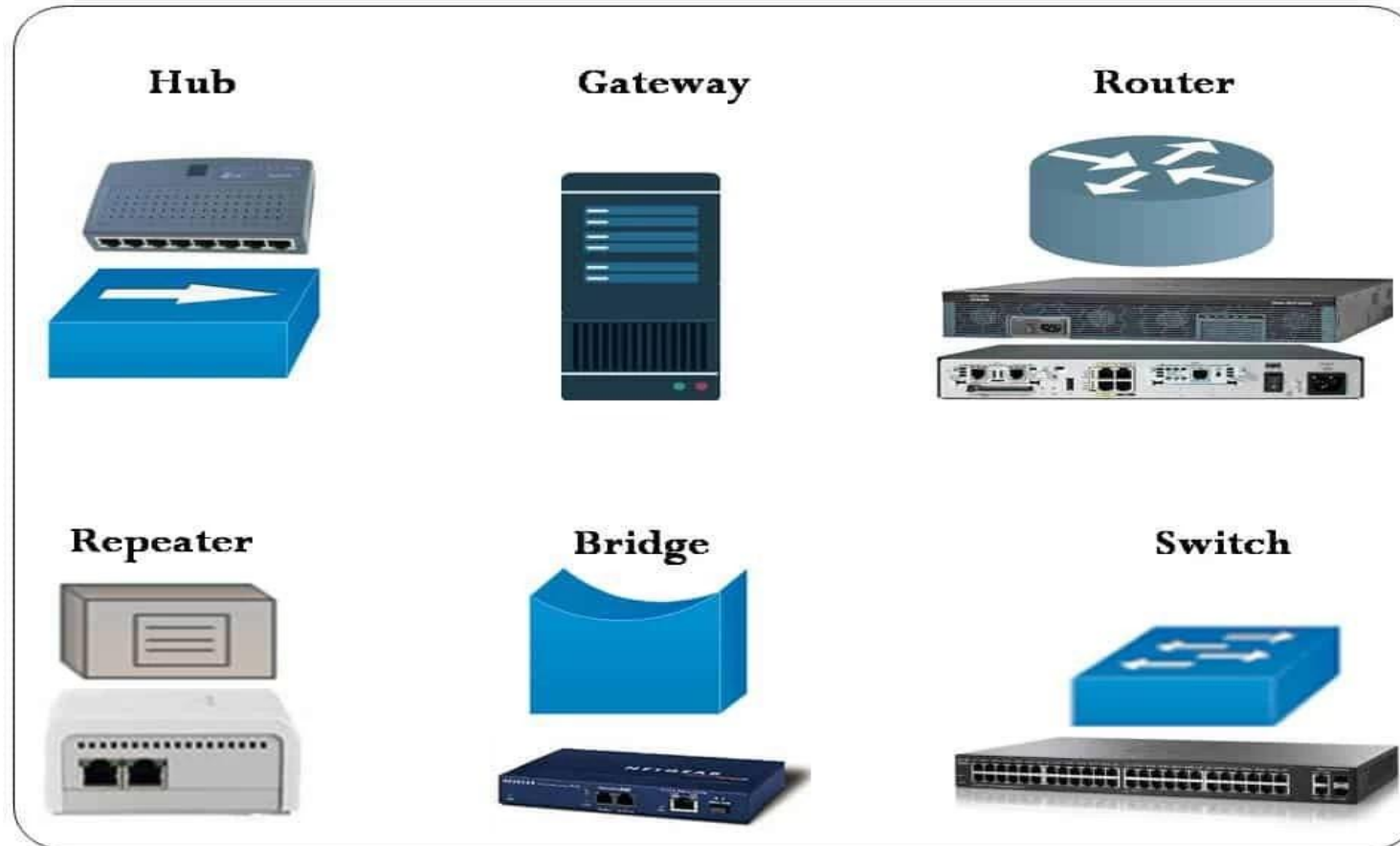
2) Software components :

- The **applications** needed to **execute** on the hardware devices are the software components of a computer network.



Basic Components - Hardware

Hardware Components



Basic Components - Hardware

Hardware Components

- **Hub** – electronic device (with a number of ports) used in a LAN to link groups of computers
- **Switch** - networking device that connects devices on a computer network by using packet switching to receive, process, and forward data
- **Repeaters** (also called amplifiers) – electronic devices that receive signals and amplify and send them along the network
- **Routers** - electronic devices used to ensure messages are sent to their intended destinations
- **Gateway** – consists of hardware and/or software that allows communications between dissimilar networks
- **Bridge** – consists of hardware and/or software that allows communication between two similar networks

Basic Components - Hardware

HUB

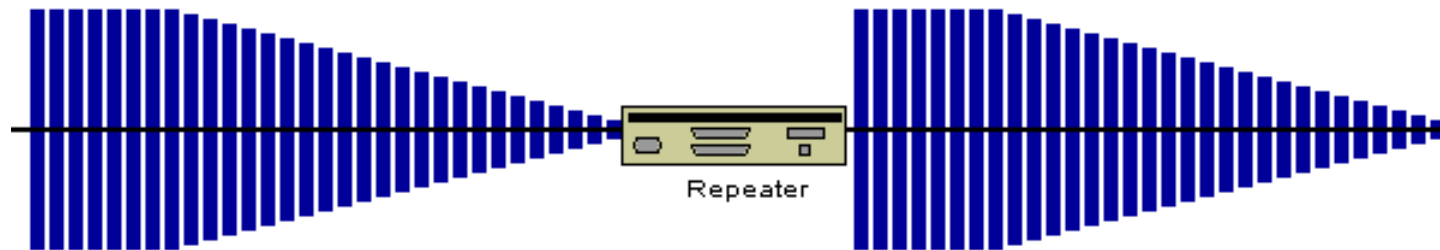
- The original Ethernet LANs relied on **certain physical characteristics** of the cable carrying the network data (usually coaxial cable).
- **New Ethernet technologies** introduced new cable types that we unable to fulfill the original physical requirements.
- **New devices - hubs** - were introduced to simulate those characteristics.
- Simply put, the hub's major function is to replicate data it receives from one device attached to it to all others.



Basic Components - Hardware

REPEATERS

- In networking, electrical signals weaken due to a wire's resistance, limiting cable length.
- A repeater overcomes this by **amplifying and retransmitting signals** between two cable segments, enabling longer connections.
- Any electrical signal reaching the **repeater from one segment**, will be amplified and retransmitted to the other segment.



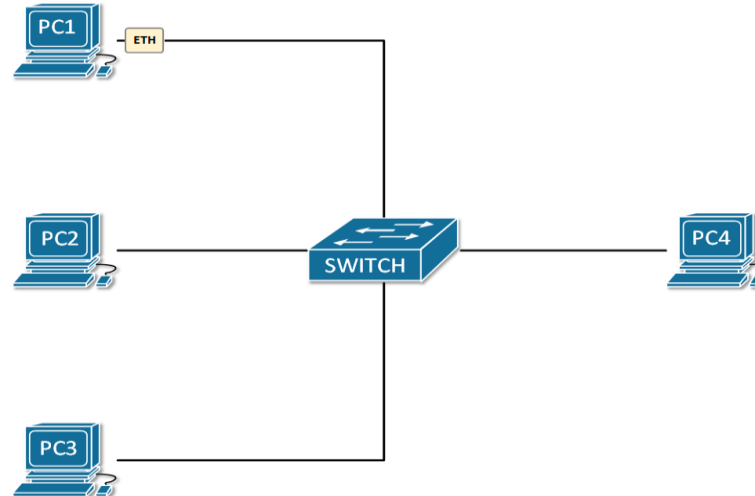
An illustration of a repeater at work

The electrical signal entering the repeater at one end is weakened.
 The repeater amplifies the electrical signals and resends the data.

Basic Components - Hardware

SWITCH

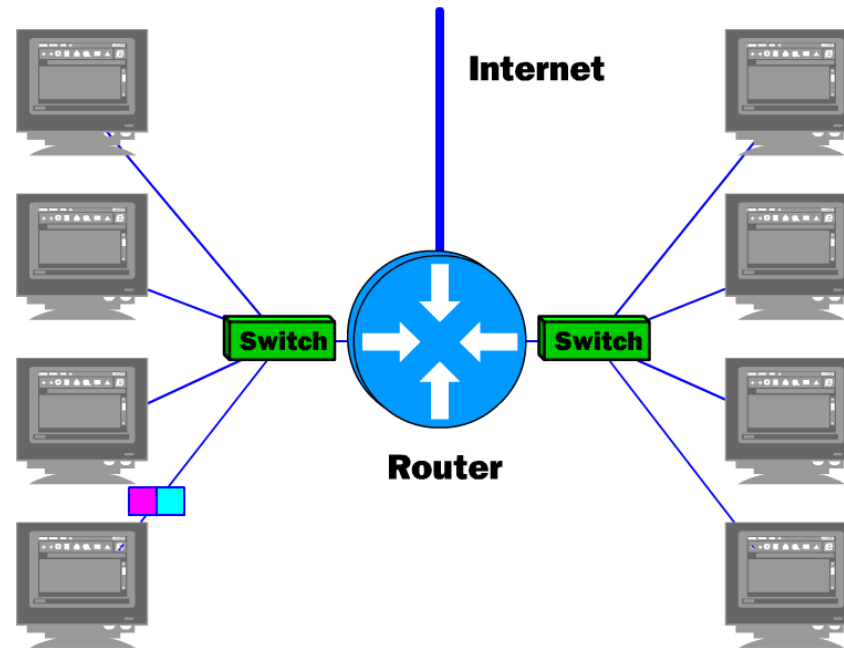
- A switch is a hardware device in a network that connects and helps **multiple devices share a network** without their data interfering with each other.
- A switch works like a traffic cop at a busy intersection. When a data packet arrives, the switch decides where it needs to **go and sends** it through the **right port**.
- Some data packets come from devices **directly connected** to the switch, like computers or VoIP phones. Other packets come from **devices connected through hubs** or routers.



Basic Components - Hardware

ROUTERS

- Routers connect **two or more networks** and **forward data packets** between them.
- When data arrives from one of the segments, the **router decides, according to its routing table**, to which segment to forward that data.



Basic Components - Hardware

GATEWAY

- "Gateway" is a term that was once used to refer to a **routing device**. Today, in the TCP/IP world, the term "router" is used to describe such a device.
- The term gateway now refers to special-purpose devices, that perform **protocol conversions**.
- Gateways implement application layer conversions of information received from various protocols.

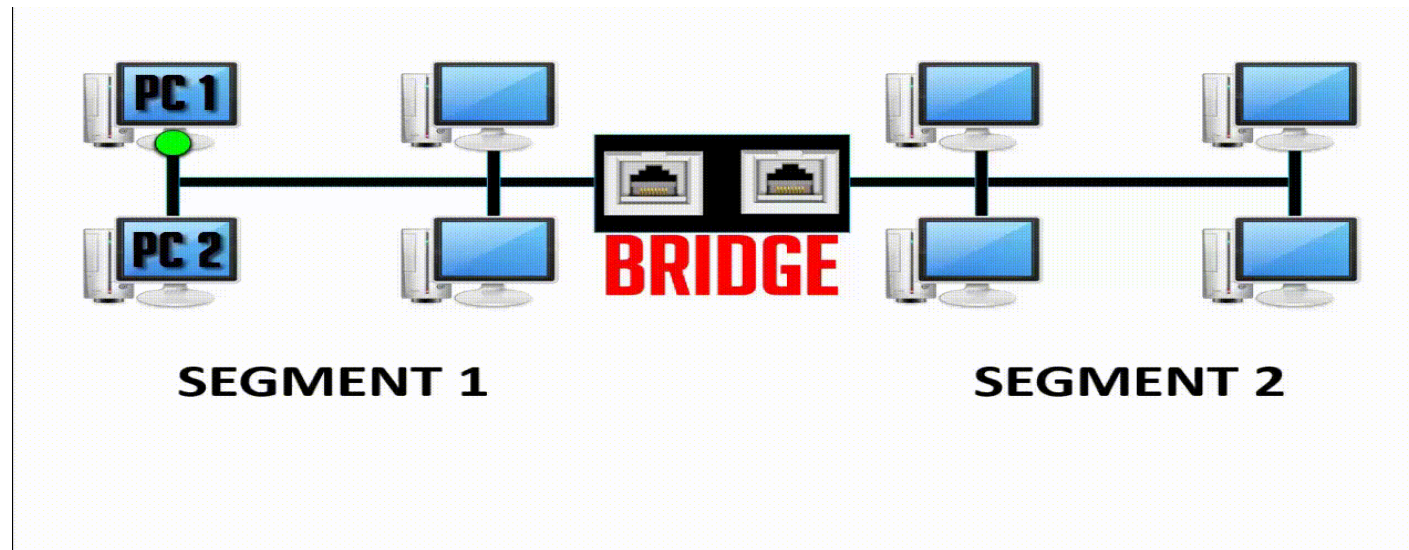


- Examples of gateways found on today's markets are:
 - 1) VocalTec Gateway:** A gateway that converts human speech traveling on analog phone lines into local area network protocol data, and visa-versa.
 - 2) RadVision Gateway:** Converts video from digital phone lines into local area network protocol data, and visa-versa.

Basic Components - Hardware

BRIDGE

- Bridge is used to divide a **large network into smaller segments**.
- A Bridge can connect **two different types of media** or network architecture, but it cannot connect two different types of network layer protocol such as TCP/IP or IPX.
- Bridge requires the same network layer protocol **in all segments**.



Basic Components - Hardware

Feature	Router	Bridge	Gateway
Function	Directs traffic between networks	Connects network segments	Translates between protocols
Layer	Network (Layer 3)	Data Link (Layer 2)	Multiple layers (4-7)
Addressing	IP addresses	MAC addresses	Depends on protocol
Use Case	Connecting different networks (e.g., LAN to WAN)	Extending a LAN	Connecting different protocols (e.g., TCP/IP to Voice over Internet Protocol (VoIP))
Example	Home router connecting to the internet	Network bridge connecting wired and wireless LANs	VoIP gateway for converting phone calls

Basic Components - Software

The software layer ensures communication and data transfer happen smoothly and securely:

- **Operating Systems:** Network Operating Systems (NOS) like Windows Server or Linux handle network requests and management.
- **Network Protocols:** A set of rules that governs data communication. Common protocols include:
 - TCP/IP: For data transmission across the internet.
 - HTTP/HTTPS: For accessing web pages.
 - FTP: For file transfers.
 - SMTP: For sending emails.
- **Firewall and Security Software:** Protects the network from unauthorized access and cyberattacks.
- **DNS (Domain Name System):** Translates human-readable domain names (e.g., google.com) into IP addresses.

How are networks categorized?

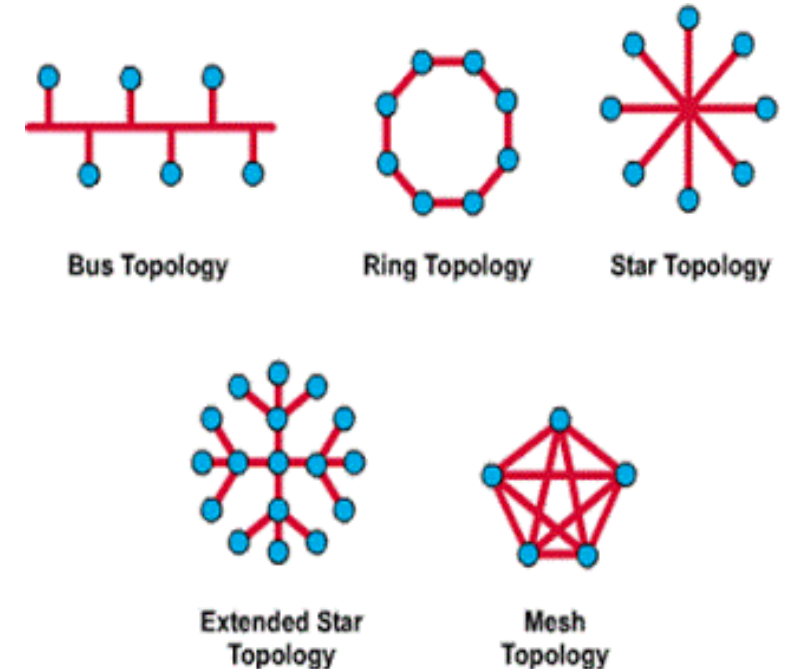
Networks are typically classified based on **three key properties**:

1. **Topology** – The physical or logical arrangement of network devices and connections.
2. **Protocol** – The set of rules governing data transmission and communication.
3. **Architecture** – The structural framework that defines how network components interact and operate.

Categorization - Network Topologies

Network Topology (or layout) – refers to the way **computers and peripherals** are configured to **form networks**.

- **Bus topology** – all computers are linked by a single line of cable
- **Star topology** – multiple computers and peripheral devices are linked to a central computer, called a host
- **Ring topology** – each computer or workstation is connected to two other computers, with the entire network forming a circle
- **Hybrid topology** – combine network layout types to meet their needs



Comparison of Network Topologies

Feature	Bus	Star	Ring	Hybrid
Structure	Single backbone cable	Central hub with connected nodes	Circular connection	Combination of multiple topologies
Cost	Low	Moderate	Moderate	High
Reliability	Low (single failure disrupts all)	High (hub failure affects all)	Moderate (one failure can disrupt)	Very High (backup connections)
Performance	Slows down with more devices	High	Moderate	Optimized for performance
Scalability	Limited	Easy to expand	Difficult	Highly scalable

Categorization – Networking Protocols

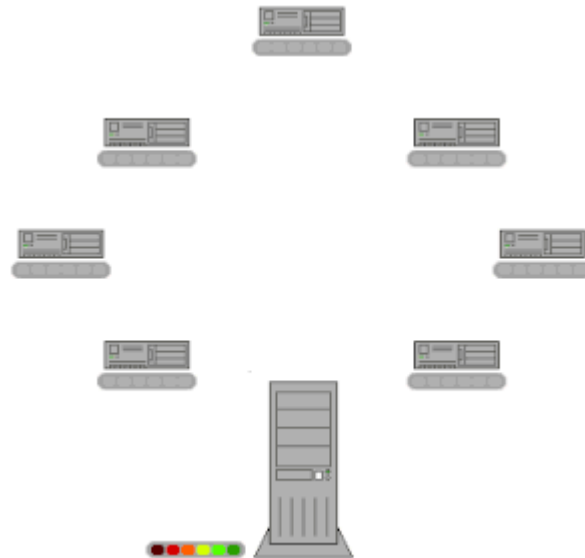
- A protocol is simply **an agreed-on set of rules and procedures** for transmitting data between two or more devices.
- **Features determined by the protocol are:**
 - How the sending device indicates it has **finished** sending the message.
 - How the receiving device indicates it has **received** the message.
 - The **type of error** checking to be used.

Categorization – Networking Protocols

- Most networks use **Ethernet**, but some may use IBM's Token Ring protocol.
- On the Internet, the major protocol is **TCP/IP** (Transmission Control Protocol/Internet Protocol), which ensures reliable data transmission across networks.
- TCP manages **data flow and error checking**, while IP handles **addressing and routing**.
- Other protocols, like **UDP** (User Datagram Protocol), provide faster, connectionless communication.
- Additionally, **wireless networks** often use Wi-Fi, which follows IEEE 802.11 standards to enable seamless connectivity.

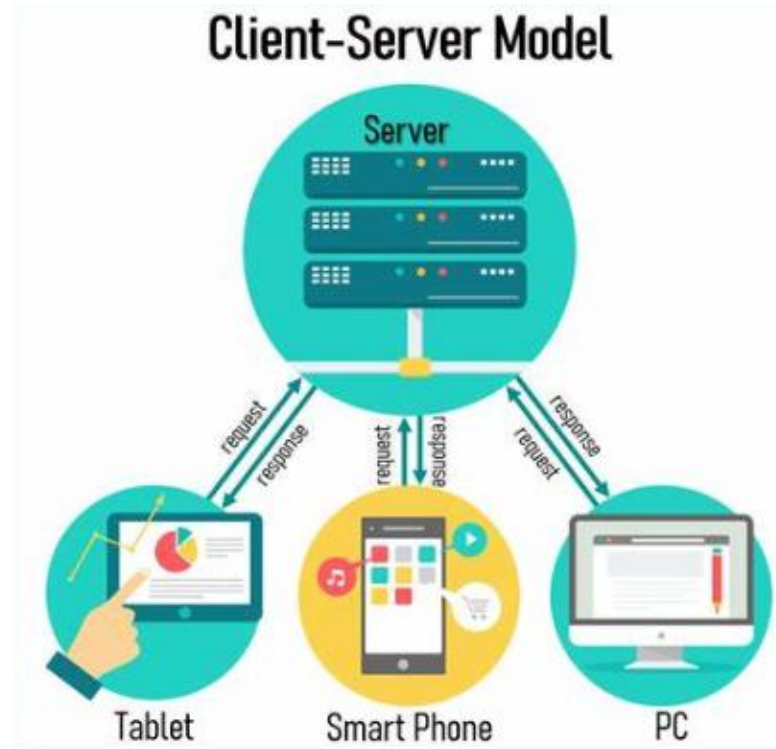
Categorization – Network Architecture

- Network architecture – refers to the way a network is **designed and built**.
- The two major types are:
 - 1) Peer-to-peer architecture** – Computers **connect with each other** in a workgroup to share files, printers, and Internet access. This is used to connect a **small number of computers**.



Categorization – Network Architecture

2) **Client/server architecture** – sends information from a **client computer to a server**, which then **relays** the **information** back to the client computer, or to other computers on the network.



OSI Layers

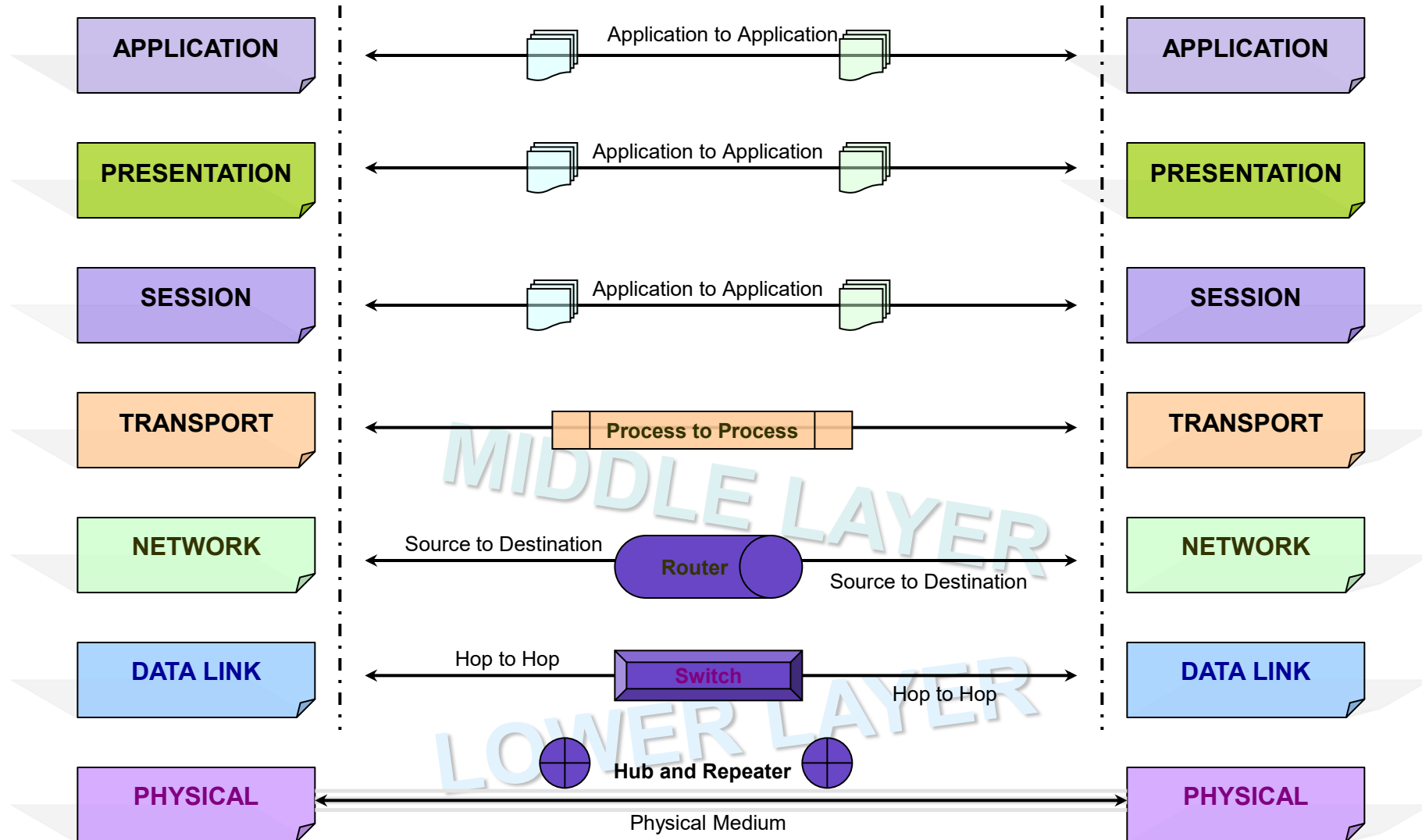


Introduction

- The Open Systems Interconnection (OSI) model is a **standard "reference model"** created by the **International Organization for Standardization (ISO)** to describe how the different software and hardware components involved in a **network communication** interact with one another.
- In its most basic form, it divides **network architecture into seven layers** which, from top to bottom, are the **Application, Presentation, Session, Transport, Network, Data-Link, and Physical Layers**.
- It is therefore often referred to as the **OSI Seven Layer Model**.

OSI Model

OSI Model's 7 Layers



OSI Model

Data, Protocol & Activities

OSI Layers	TCP/IP Suit and Protocols	Activities
Application	Application Telnet, FTP, SMTP, HTTP, DNS, SNMP, <i>Specific address</i> etc...	To allow access to network resources
Presentation	Presentation	To Translate, encrypt, and compress data
Session	Session	To establish, manage, and terminate session
Transport	Transport SCTP, TCP, UDP, Sockets and <i>Ports address</i>	To Provide reliable process-to-process Message delivery and error recovery
Network	Network IP, ARP/RARP, ICMP, IGMP, <i>Logical address</i>	To move packets from source to destination; to provide internetworking
Data Link	Data Link IEEE 802 Standards, TR, FDDI, PPP, <i>Physical address</i>	To organize bits into frames; to provide Hop-to-hop delivery
Physical	Physical Medium, Coax, Fiber, 10base, Wireless	To Transmit bits over a medium; to provide Mechanical and electrical specifications

Physical Layer



Physical Layer

Introduction

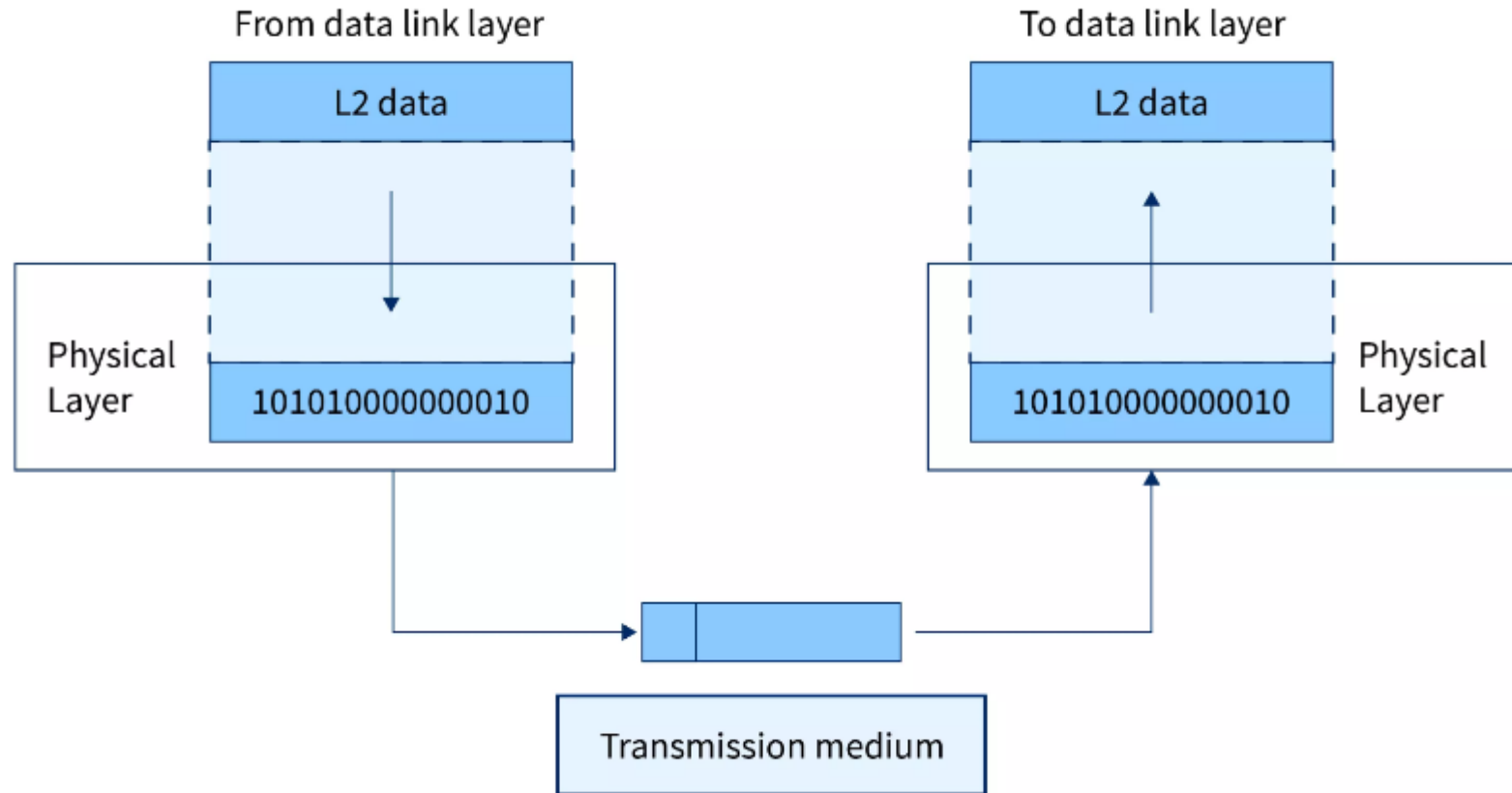
- The physical layer is the **lowest layer** of the OSI model.
- It handles the **transmission of raw bitstreams** over the physical medium.
- It is responsible for **activating, maintaining, and deactivating** physical connections.
- This layer sends and receives **unstructured bits (0s and 1s)**.
- It **encodes bits into signals** for transmission.
- The physical layer also **sets voltages, light speed** (in fiber optics), and **data rates** (bits per second).

Physical Layer - Functions

- **Bit Synchronization:** Uses a **clock** to sync **sender and receiver** at the bit level.
- **Bit Rate Control:** Sets the **number of bits** sent per second (data rate).
- **Physical Topologies:** Defines **device arrangement** like **bus, star, or mesh**.
- **Transmission Mode:** Controls data flow as **Simplex, Half-Duplex, or Full-Duplex**.

Physical Layer

Physical Layer - Data Transmission and Working

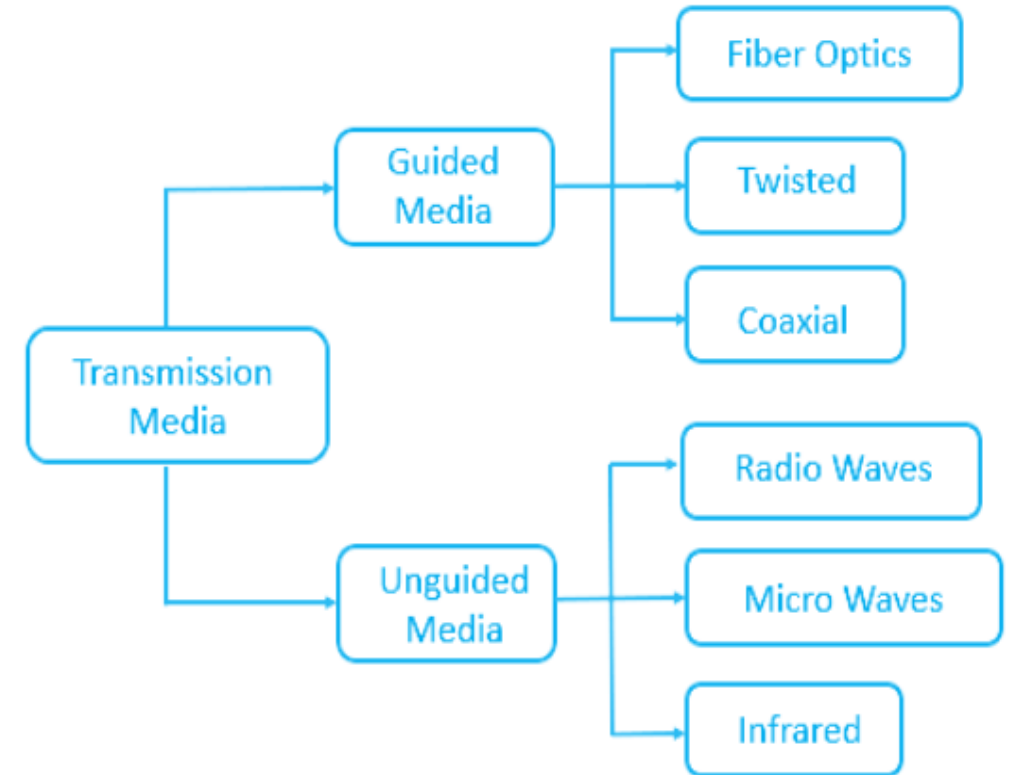


- *Hub, Repeater, Modem, and Cables are Physical Layer devices.*
- *Network Layer, Data Link Layer, and Physical Layer are also known as **Lower Layers** or **Hardware Layers**.*

Physical Layer

Transmission Medium

- Transmission media refer to the **physical pathways** through which data is transmitted from **one device to another within a network**.
- It is a physical way for **electromagnetic signals** to transport data.
- These pathways can be **wired or wireless**.
- The choice of medium depends on factors like **distance, speed, and interference**.



Physical Layer

Transmission Medium – Guided Medium

- **Twisted Pair Cable**
 - Most common in Ethernet networks
 - Copper wires twisted to reduce interference
- **Coaxial Cable**
 - Copper core with shielding
 - Supports high-speed data over long distances
- **Fiber Optic Cable**
 - Uses glass/plastic to send data as light
 - Very fast, long-distance, no EMI issues



Copper Twisted pair



Copper Coaxial Cable



Optic Fiber Cable

Physical Layer

Transmission Medium – Unguided Medium

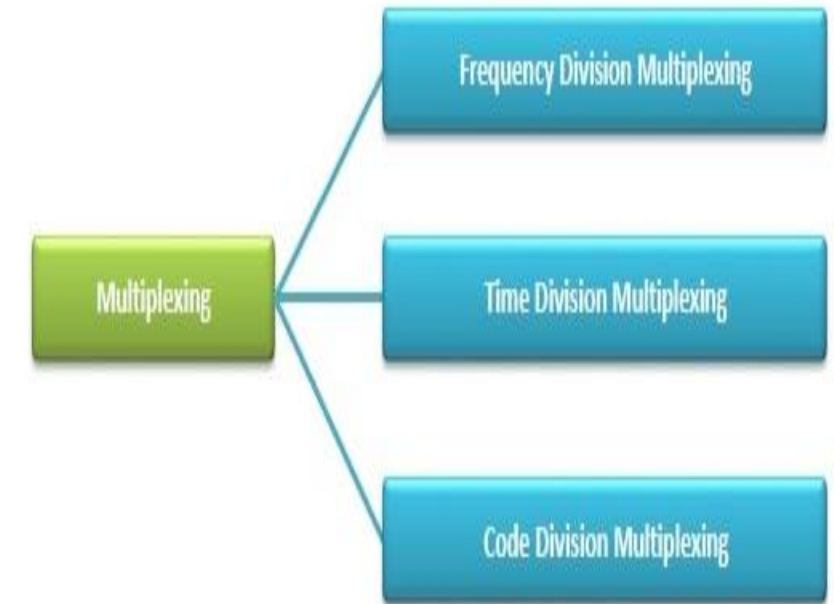
- **Radio Waves**
 - Most common wireless medium
 - Uses radio frequencies to send data
- **Microwaves**
 - High-frequency waves
 - Used for long-distance line-of-sight communication
- **Infrared Waves**
 - Uses infrared light
 - Short-range use (e.g., remotes, mobile-to-printer)



Physical Layer

Multiplexing

- Multiplexing is a technique used to **combine multiple signals** and send them over a **single communication channel**.
- It allows the efficient use of the channel by **sharing it among multiple data streams**.
- Each data stream gets a **logical portion** of the channel's capacity.
- **Example** : Imagine multiple people talking over a single phone line, each taking turns very quickly — the receiver can separate their voices properly.



Data Link Layer



Data Link Layer

Role of Data Link Layer

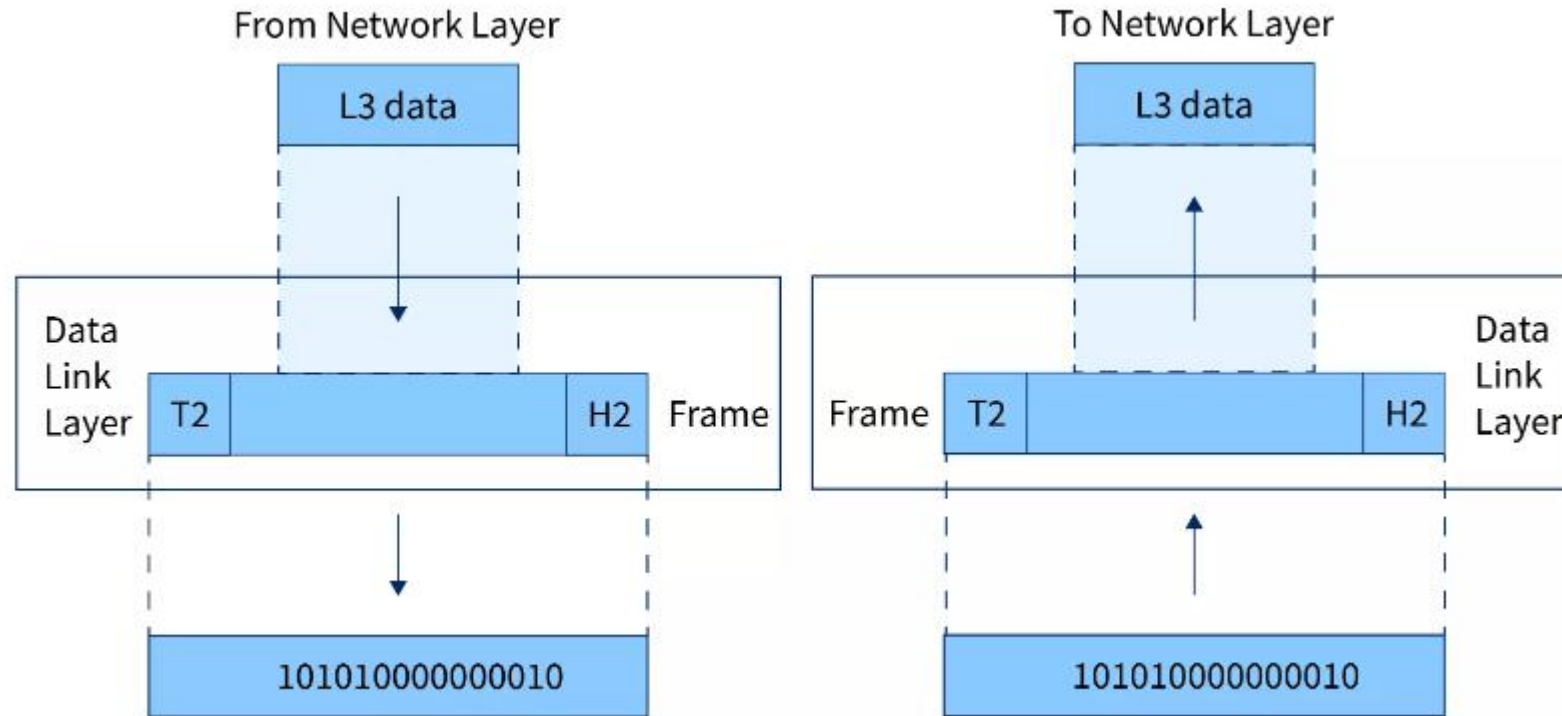
- The **Data Link Layer** is the second layer in the **OSI (Open Systems Interconnection) model**. It plays a crucial role in enabling communication between devices in a network by **ensuring error-free data transfer and proper addressing**.

Key Functions of the Data Link Layer:

- **Framing**: Converts raw bits into structured frames for transmission.
- **Addressing**: Uses MAC (Media Access Control) addresses to identify devices in a local network.
- **Error Detection & Correction**: Uses techniques like Cyclic Redundancy Check (CRC) to detect errors.
- **Flow Control**: Ensures that a fast sender does not overwhelm a slow receiver.
- **Access Control**: Determines how multiple devices share the same communication channel (via MAC sublayer).

Data Link Layer

Data Link Layer - Data Transmission and Working



- Packet in the Data Link layer is referred to as **Frame**.
- Data Link layer is handled by the NIC (Network Interface Card) and device drivers of host machines.
- Switch & Bridge are Data Link Layer devices.

Data Link Layer

MAC (Media Access Control Address)

- A MAC address is a **unique identifier** assigned to a **Network Interface Card (NIC)** of a device. It operates at the Data Link Layer and is used for **local addressing within a LAN**.

Characteristics of a MAC Address:

- **48-bit address** (expressed in hexadecimal format, e.g., 00:1A:2B:3C:4D:5E).
- **Globally unique** (assigned by the manufacturer).
- **Used in Ethernet, Wi-Fi, and Bluetooth networks.**
- Divided into **two parts**:
 - **OUI** (Organizationally Unique Identifier) - **First 24 bits**: Identifies the manufacturer.
 - **NIC-Specific Identifier** - **Last 24 bits**: Uniquely identifies the device.

A MAC (Media Access Control) address is also called a physical address because it is hardcoded into the network interface card (NIC) of a device at the time of manufacturing.

Data Link Layer

MAC (Media Access Control Address)

```
Command Prompt
Microsoft Windows [Version 10.0.22631.4890]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Mani>getmac

Physical Address      Transport Name
=====
00-45-E2-6A-A0-8B     \Device\Tcpip_{F983D3A5-AC0F-41C5-ACDD-D2DB4F3B1D48}
00-45-E2-6A-A0-8C     Media disconnected
54-8A-F9-E9-94-03     Disconnected

C:\Users\Mani>
```

Data Link Layer

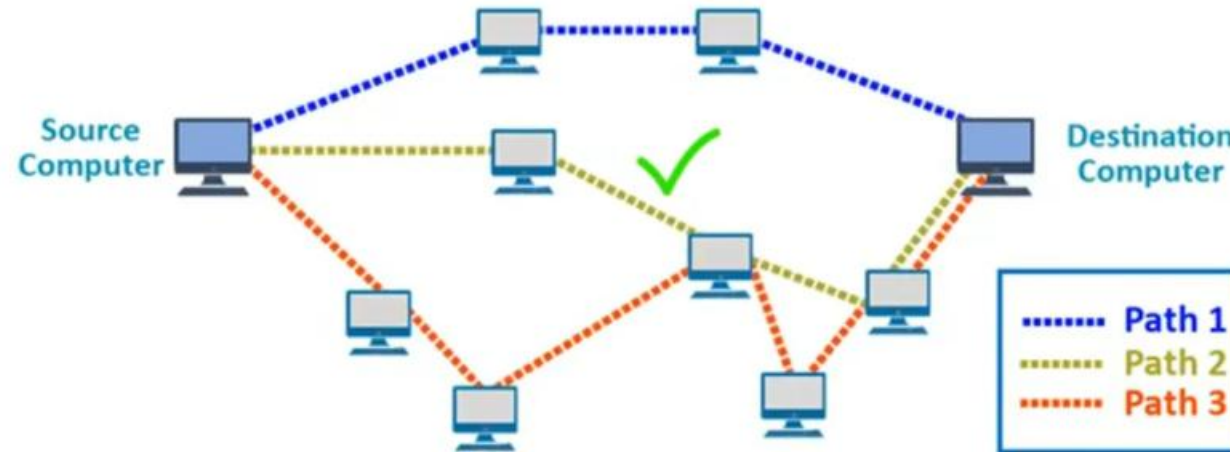
Relation Between Data Link Layer & MAC Address

- The Data Link Layer ensures data is transmitted correctly between devices within a local network.
- The MAC address helps identify the source and destination of the data frame within the same network.
- Devices in a network communicate using MAC addresses when forwarding data at Layer 2.
- **Example Scenario:** If a computer **sends a packet to another computer** in the same LAN
 1. The sender encapsulates the data with the destination MAC address.
 2. The switch reads the MAC address and forwards the frame to the correct device.
 3. The receiver extracts the data and processes it.

Data Link Layer

Switching

- Switching is the process of **forwarding data** from one device to another within a network.
- In a computer network where many different computers are connected to each other, there can be **more than one path** to send data from one computer to another.
- **Selecting a path** that data must take out of the available options is called switching.
- Network switches operate at the **data link layer (Layer 2) of the OSI model**, accepting packets from connectec



Why is Switching Required?

- Switching is essential in networking to ensure **efficient, fast, and reliable communication** between devices. Here are some **key reasons** why switching is essential:

1. Efficient Data Transmission

- Switching enables **direct communication** between devices in a network, avoiding unnecessary data collisions. Ensures that data reaches the intended recipient **without flooding the entire network**.

2. Reduces Network Congestion

- Switches create **dedicated paths** between devices, reducing network traffic and congestion.
- Unlike hubs (which broadcast data to all devices), switches forward data only to the **intended destination**.

3. Scalability & Flexibility

- Switching enables **scalable networks** by allowing more devices to connect without performance degradation. Used in **small LANs to large enterprise networks** with thousands of devices.

Types of Switching

- There are **three main types** of switching:

1. Circuit Switching

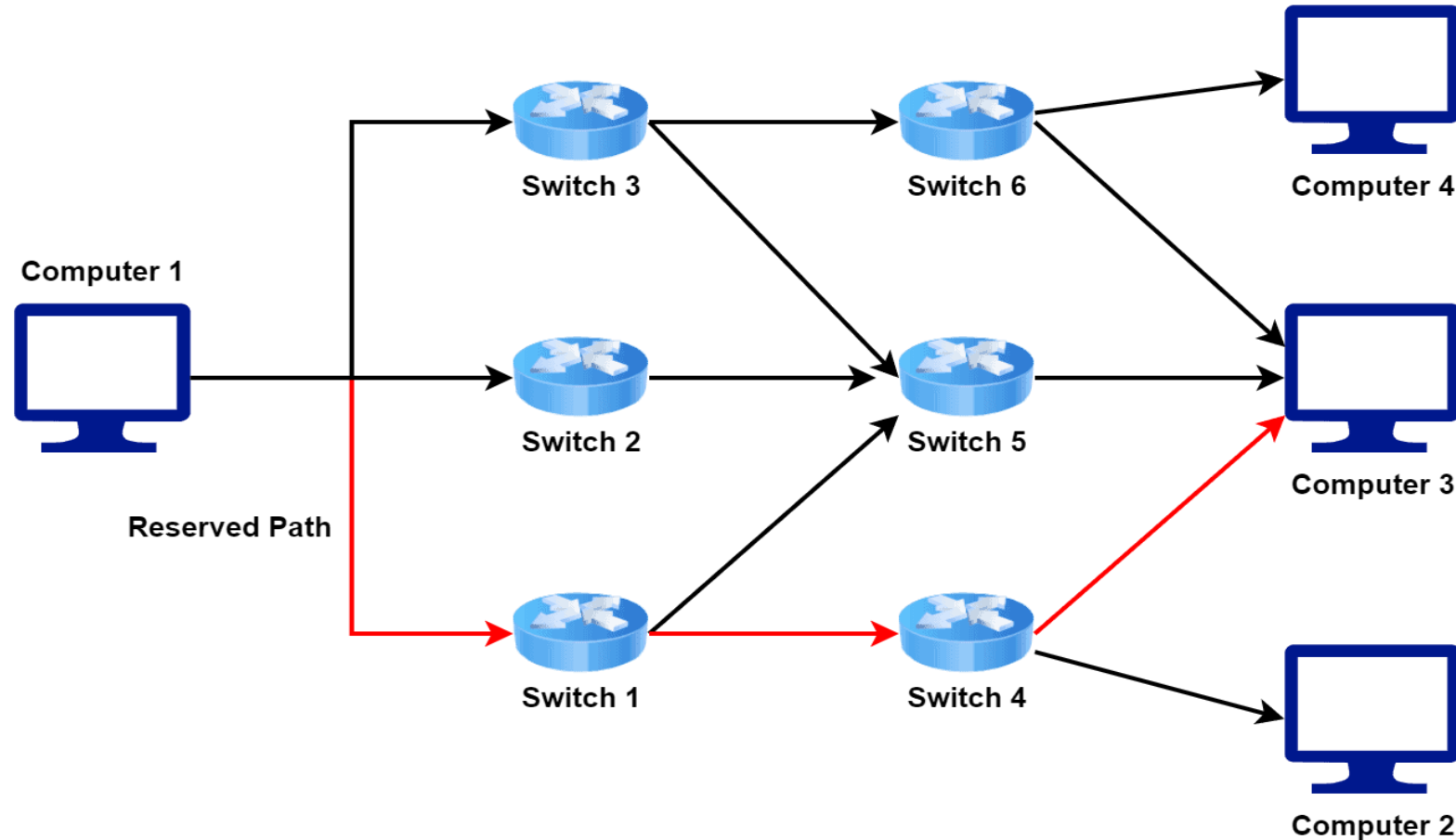
- Establishes a **dedicated communication path** between sender and receiver before data transmission.
- **Example:** Traditional telephone networks.
- **Pros:** Reliable, no data loss.
- **Cons:** Inefficient for data networks as resources remain occupied even when no data is transmitted.

Circuit switching is **inefficient for data networks** because it reserves bandwidth even when unused, making it **hard to scale in high-traffic areas**. However, it is still used in **satellite communications** where low delay is essential.

Data Link Layer

Types of Switching

Example of Circuit Switching



Types of Switching

- There are **three main types** of switching:

2. Packet Switching

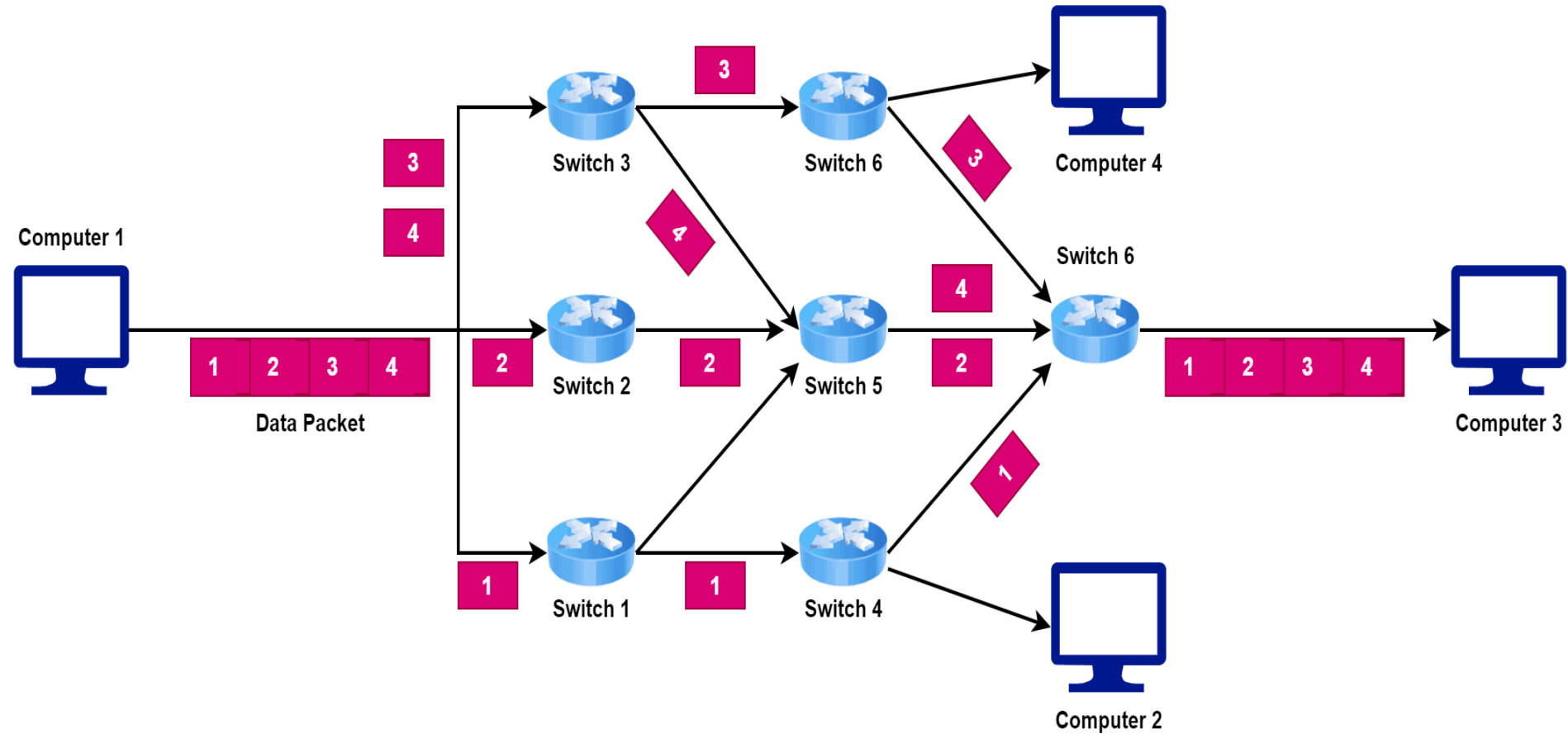
- Data is **broken into packets** and sent independently through the network
- **Example:** Internet communication.
- **Types:**
 - **Connectionless** (Datagram Switching) – Each packet takes a different path (e.g., UDP).
 - **Connection-Oriented** (Virtual Circuit Switching) – A predefined route is established (e.g., TCP).
- **Pros:** Efficient, scalable.
- **Cons:** Packets may arrive out of order, requiring reassembly.

In packet switching, each packet travels separately and may take different paths to reach its destination. This allows **multiple packets to be sent at the same time**, making better use of the network.

Data Link Layer

Types of Switching

Example of Packet Switching



Types of Switching

- There are **three main types** of switching:

3. Message Switching

- The entire message is stored and forwarded **hop-by-hop**.
- No dedicated path; each node **stores the message** before forwarding it.
- **Example:** Early email systems.
- **Pros:** No need for a dedicated channel.
- **Cons:** High latency, requires large storage at intermediate nodes.

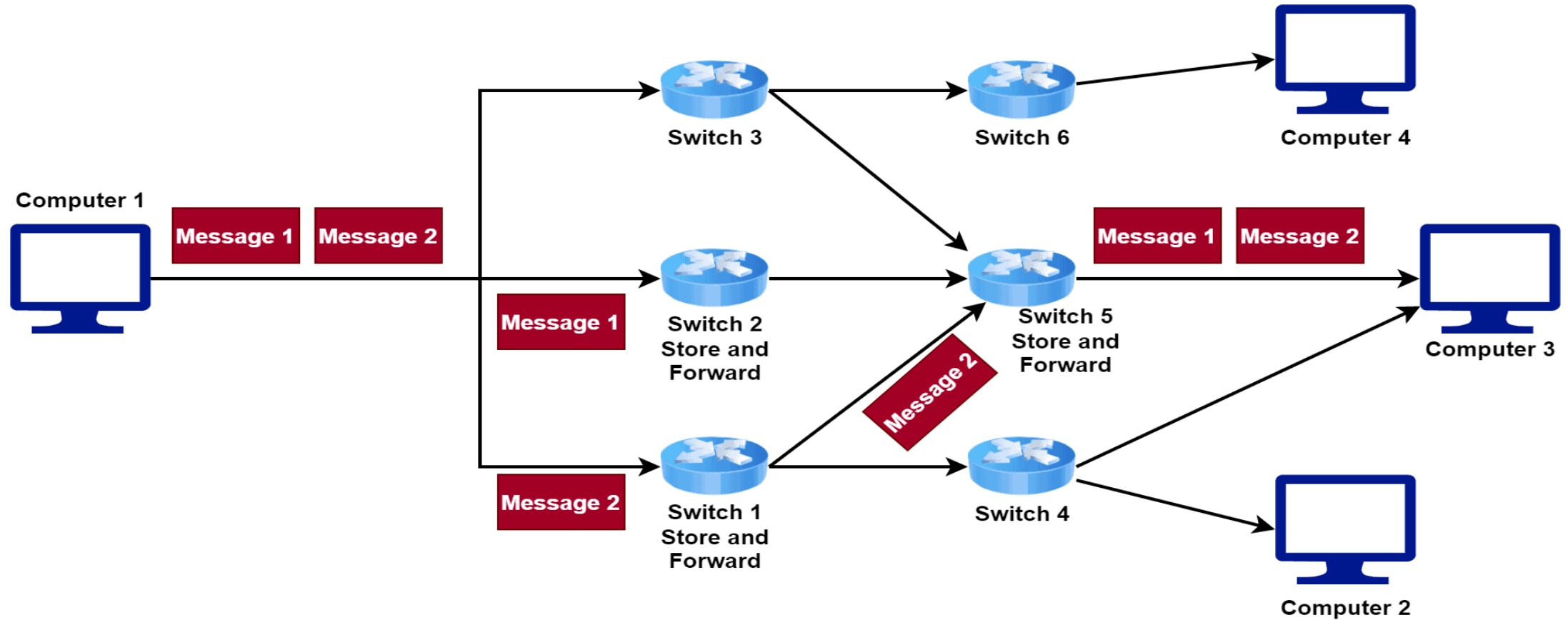
Hop-by-hop means data packets **travel from one network device (like a router or switch) to another** until they reach their destination. Each device decides where to send the packet next using its routing table.

This method helps **route packets efficiently** and allows them to take different paths if needed, making the network more reliable and flexible.

Data Link Layer

Types of Switching

Example of Message Switching



Types of Switching - Comparison

Feature	Circuit Switching	Message Switching	Virtual Packet Switching	Datagram Packet Switching
Connection Type	Connection-Oriented	Connection-Less	Connection-Oriented	Connection-Less
Path & Resource Allocation	Follows Dedicated Path and Blocks Resources	No dedicated path, needs storage at each switch	Follows Dedicated Path and Blocks Resources	No dedicated path, needs storage at each switch
Data Transmission	Entire message is sent together	Entire message is sent together	Message is broken down into packets	Message is broken down into packets
Traffic Impact	Causes Traffic Congestion	Reduces Traffic Congestion	Can Cause Traffic Congestion	Reduces Traffic Congestion
Data Loss	No loss of information	No loss of information	No loss of information	Packets can get lost
Cost & Complexity	Expensive + Easy	Cost Efficient + Easy	Expensive + Complex	Cost Efficient + Easy

Ethernet and VLAN Concepts

1. Ethernet

- A wired **networking technology** used in LANs for data transmission.
- Uses **MAC addresses** for communication.
- Operates at **Layer 2 (Data Link Layer)** of the OSI model.

2. VLAN (Virtual Local Area Network)

- A VLAN logically divides a **physical network** into **multiple isolated networks**.
- **Purpose:** Enhances security, reduces congestion, and improves network management.
- **Example:**
 - VLAN 10 → HR Department
 - VLAN 20 → IT Department
 - VLAN 30 → Guest Network

By using VLANs, organizations can **isolate traffic** without requiring separate physical switches.

Data Link Layer

Trunking

- Trunking is a **networking method** used to **carry multiple VLANs over a single physical link** between network devices like switches, routers, or servers.
- It is widely **used in enterprise networks** to **improve efficiency, reduce cabling**, and enable VLAN communication across multiple network segments.

Why is Trunking Needed?

- **Reduces the number of physical cables** required to connect switches.
- Ensures **efficient VLAN communication** between different switches.
- Helps in **scalability**, allowing VLANs to span across the network.
- Used in **data centers and enterprise networks** where multiple VLANs exist.

Flow Control

- Flow control is a set of **procedures** that restrict the **amount of data a sender** should send before it waits for some **acknowledgment** from the receiver.
- The **two types of Flow Control protocols** shown in the Data Link Layer are:
 1. **Stop-and-Wait Protocol** :A simple flow control technique where the sender sends **one frame at a time** and waits for an **acknowledgment** before sending the next.
 2. **Sliding Window Protocol** : A more efficient protocol used in noisy channels, where the sender can send **multiple frames** before needing an acknowledgment, allowing **better utilization** of the channel.

Error Control

- Error Control is a **combination** of both **error detection and error correction**. It ensures that the data received at the receiver end is the same as the one sent by the sender.
- Error detection is the process by which the **receiver informs the sender** about any erroneous frame (damaged or lost) sent during transmission.
- Error correction refers to the **retransmission** of those frames by the sender. error-correction-in-link-layer.
- **Error control methods** are mainly divided into:
 1. **Stop-and-Wait ARQ(Automatic Repeat reQuest.)**
 2. **Sliding Window ARQ**
 - 2.1 **Go-Back-N ARQ**
 - 2.2 **Selective Repeat ARQ**

Error Control Methods

ARQ - Automatic Repeat reQuest

- ARQ is an **error control protocol** used in data communication.
- It ensures reliable data transmission over **unreliable or noisy communication** channels by using acknowledgments (ACKs) and retransmissions.

How ARQ Works:

- **Sender sends a frame.**
- **Receiver checks** the frame for errors. If the frame is correct, the receiver sends an **ACK** (acknowledgment).
- If the frame is **damaged or lost**, the receiver sends a **NACK** (negative acknowledgment) or doesn't respond.
- The sender **retransmits** the frame after a timeout or on receiving a NACK.

Error Control Methods

1. Stop-and-Wait ARQ

- Sends **one frame at a time**
- Waits for **acknowledgment (ACK)** before sending the next
- Simple, but **inefficient for high-speed networks**

2. Sliding Window ARQ

- Allows sending **multiple frames** before waiting for ACKs
- Uses a window size to control the **number of unacknowledged frames**
- Efficient and supports **continuous transmission**

Error Control Methods

2.1 Go-Back-N ARQ

- If a frame is lost or damaged, **all frames** from that point are **resent**
- Sender goes “**back**” to the lost frame
- Simple but can cause **redundant retransmissions**

2.2 Selective Repeat ARQ

- Only the **incorrect/lost frames** are retransmitted
- **Correct frames** are **buffered**
- More **efficient** but **complex** to implement

Error Detection Techniques

Three methods are used to detect an error in frames:

1. **Parity Check:** Adds a parity bit to ensure the total number of 1s is **even or odd**. It's simple but can only **detect single-bit errors**.
2. **Checksum:** Data is divided into **segments, summed, and the result** (checksum) is sent with the data.
The receiver verifies it to **detect errors**. Used in **TCP/IP**.
3. **CRC (Cyclic Redundancy Check):** Divides data by a **generator polynomial** and sends the remainder (**CRC bits**) with data. It's **highly accurate**, detecting burst errors.

Data Link Layer

Access Control

- When **multiple devices** share a communication channel, the Data Link Layer (DLL) manages who gets to use the channel to **avoid collisions**.
- To manage channel access and **prevent collisions**, several MAC (Media Access Control) **methods** are used, including:
 - **CSMA** (Carrier Sense Multiple Access)
 - **CSMA/CD** (Collision Detection)
 - **CSMA/CA** (Collision Avoidance)

Access Control

- **CSMA (Carrier Sense Multiple Access)** : Devices first **sense the channel** before transmitting. If it's idle, they send data; if busy, they wait.
- **CSMA/CD (Collision Detection)** : Used in **wired networks** (like old Ethernet). Devices listen while sending; if a **collision** is **detected**, transmission **stops** and retries after a random delay.
- **CSMA/CA (Collision Avoidance)** : Used in **wireless networks**. Devices try **to avoid collisions** before sending by using signals like RTS/CTS (Request to Send/Clear to Send).

Network Layer



Network Layer

Role of Network Layer

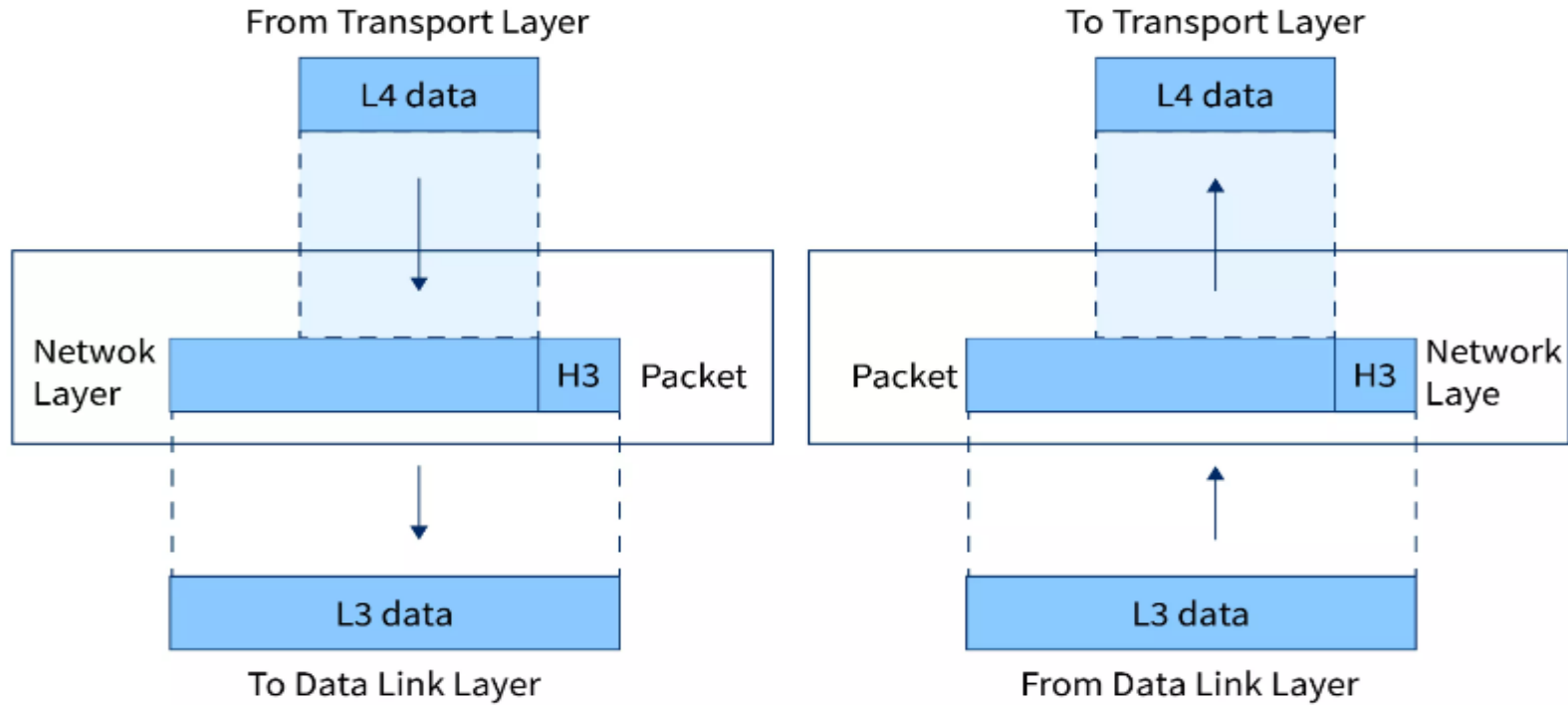
- The Network Layer (**Layer 3 of the OSI model**) is responsible for **routing and forwarding data across different networks**. It ensures that data packets reach their destination efficiently, even if they pass through **multiple networks**.

Key Functions of the Network Layer:

- **Logical Addressing (IP Addressing)**: Assigns a **unique IP address to devices** for identification and communication across networks.
- **Routing**: Determines the best path for data transmission using routing protocols
- **Packet Forwarding**: Moves data packets between networks using routers and routing tables.
- **Fragmentation & Reassembly**: Splits large packets into smaller fragments for transmission and reassembles them at the destination.
- **Error Handling & Congestion Control**: Detects and manages packet loss, delays, and congestion using ICMP for error reporting.

Network Layer

Network Layer - Data Transmission and Working



- *Segment in the Network layer is referred to as **Packet**.*
- *Network layer is implemented by networking devices such as routers and switches.*

Network Layer

Network Layer Addressing

- Network layer addressing refers to **assigning a logical address** to each device on a network to uniquely identify it for **communication** across networks.
- The most common example is the **IP address (like 192.168.1.1)**.
- It helps **route packets** from source to destination across different networks.
- Unlike MAC addresses (which are physical and fixed), network addresses can **change based on the device's location** in the network.
- **Used for:** Routing, packet delivery, and device identification in the Network Layer (Layer 3 of the OSI model).

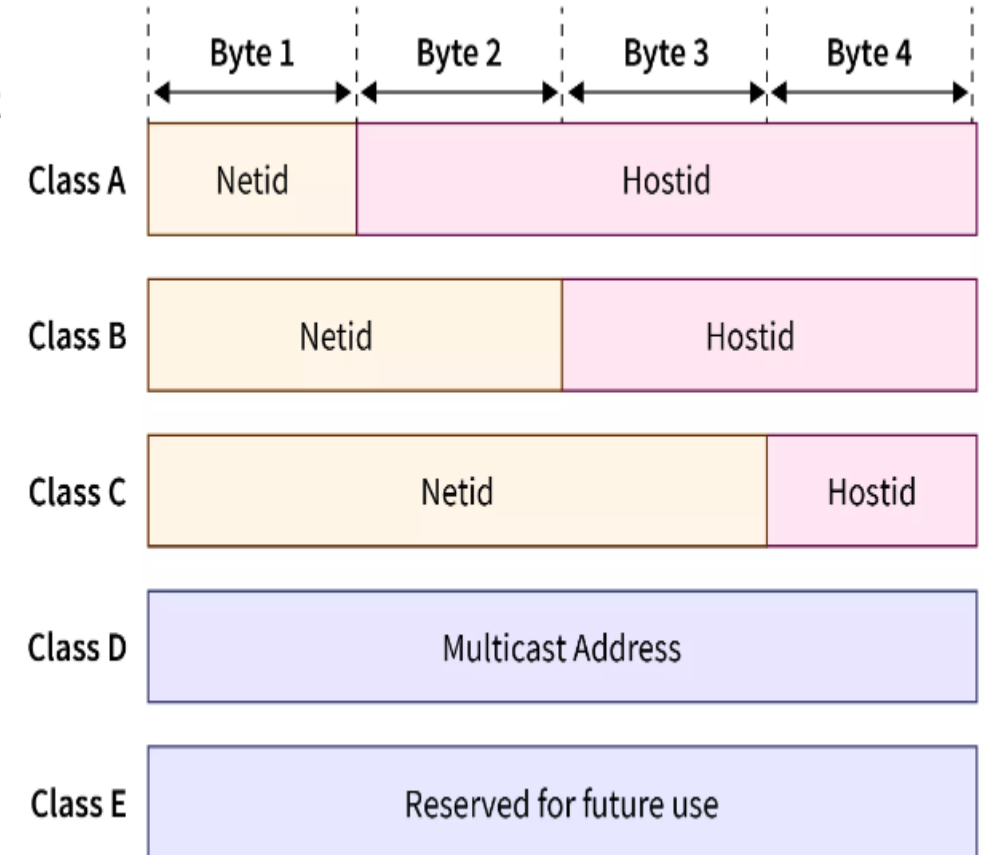
IPv4 - Network Layer Addressing

- IPv4 is the **fourth version** of the Internet Protocol, used to **identify devices** on a network using an **addressing system**.
- It uses a **32-bit address**, typically written as four decimal numbers separated by dots (**e.g., 192.168.1.1**).
- IPv4 provides about **4.3 billion unique addresses**.
- Due to limited address space, **classful addressing** was introduced to allocate IPs based on network size.

Network Layer

IPv4 – Classful Addressing

- There are 2 parts of the IPv4 address- the **network part** and the **host part**.
 - Network part is also known as the net id
 - Host part is also known as the host id.
- There are 5 types of classful addressing namely - **Class A, Class B, Class C, Class D, and Class E**.
- They are classified based on **address space** which is divided into a **fixed number of blocks** and has a **fixed number of hosts**.



Network Layer

IPv4 – Classful Addressing

Class	Starting Bits	IP Range	Default Subnet Mask	Hosts per Network	Use
A	0xxxxxxx	0.0.0.0 to 127.255.255.255	255.0.0.0	~16 million	Very large networks
B	10xxxxxx	128.0.0.0 to 191.255.255.255	255.255.0.0	~65,000	Medium networks
C	110xxxxx	192.0.0.0 to 223.255.255.255	255.255.255.0	254	Small networks
D	1110xxxx	224.0.0.0 to 239.255.255.255	N/A	N/A	Multicast
E	1111xxxx	240.0.0.0 to 255.255.255.255	N/A	N/A	Experimental use

IPv4 – Classless Addressing

- Classless Addressing (CIDR) : Classless addressing **removes the fixed size of IP** address classes (A, B, C).
- It was introduced through **CIDR (Classless Inter-Domain Routing)**.
- CIDR allows using **variable-length subnet masks** (VLSM) for better IP address utilization
- A **subnet mask** helps identify the **network portion of an IP** address **using bitwise AND**.
- **Example:**
 - IP: 192.168.10.15
 - Subnet Mask: 255.255.255.0
 - Network: 192.168.10.0
- **CIDR notation:** Written as IP/prefix, e.g., 192.168.10.0/24 (Here, /24 means the first 24 bits are the network part.)

Network Layer

IPv6

- Successor to IPv4, developed to solve the problem of **IP address exhaustion**.
- Uses **128-bit addresses (vs. 32-bit in IPv4)**, allowing vastly **more unique addresses**
- .Written in **hexadecimal**, separated by colons (e.g., 2001:0db8:85a3::8a2e:0370:7334).
- Supports features like **better security, faster routing**, and no need for **NAT**.
- Designed to **simplify address assignment** and improve **efficiency** in data handling.

Network Layer

IPv4 vs IPv6

Feature	IPv4	IPv6
Address Length	32-bit	128-bit
Address Format	Decimal (e.g., 192.168.1.1)	Hexadecimal (e.g., 2001:0db8::1)
Address Space	~4.3 billion	Vast (2^{128} addresses)
Header	Complex, includes checksum	Simplified, no checksum
Security	Optional (IPSec not built-in)	Mandatory IPSec support
NAT Required	Yes (due to address shortage)	No (unique addresses for all devices)
Configuration	Manual or DHCP	Auto-configuration supported
Broadcast	Supported	Not supported (uses multicast)

Subnetting in IP Addressing

- Subnetting divides a **large network into smaller logical sub-networks** (subnets).
- It helps in:
 - Efficient **IP address usage**
 - Reducing **network traffic**
 - Enhancing **speed and security**
 - Managing **LAN segments** without needing **extra network IDs** from the Internet Service Provider.

How Does Subnetting Work?

Steps in Subnetting (IPv4) :

1. Determine Requirements

- Know how many subnets or hosts you need.

2. Choose a Subnet Mask

- Select a subnet mask that supports your required number of subnets or hosts.
- Example: /24 (255.255.255.0), /26 (255.255.255.192), etc.

3. Calculate Subnet Details

- Use formulas:
 - Number of subnets = 2^n (n = borrowed bits)
 - Hosts per subnet = $2^h - 2$ (h = host bits)

How Does Subnetting Work?

4. Find Subnet Addresses

- Break the IP range into equal subnet blocks using the calculated subnet size.

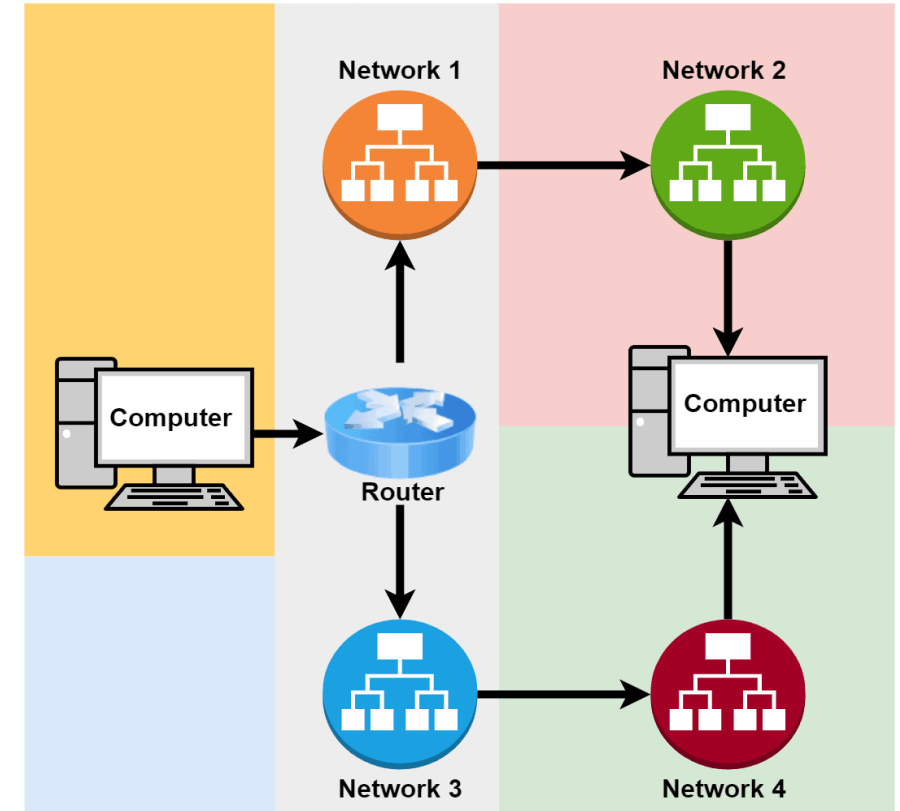
5. Assign Subnets

- Assign each subnet to different departments, devices, or networks as needed.

Network Layer

Routing

- Routing in networking is the process of **determining the best path** for data to travel from one device to another within or across networks.
- In a network, data is transmitted in the form of packets, and routing is responsible for directing these packets through multiple networks and routers, ensuring they reach their **destination efficiently and accurately**.
- It is particularly important in **large-scale networks** like the Internet, where millions of devices need to communicate across vast distances.



Why Routing is Important?

1. Path Determination

- Routing helps **find the best path** for data packets to travel from the source to the destination.
- It considers factors like **network congestion**, link **cost**, and **reliability**.

2. Inter-Network Communication

- Enables communication between devices in different networks (e.g., from a private LAN to the internet).
- Uses IP addresses to identify source and destination devices.

3. Load Balancing

- Distributes traffic across multiple paths to **avoid congestion**.
- Improves network performance and prevents bottlenecks.

4. Security & Access Control

- Firewalls and routers use routing policies to control which packets are allowed or blocked.
- Helps prevent unauthorized access and cyber threats.

Types of Routing

- Routing is categorized based on **how routes are determined and managed in a network**. The main types are:
 1. **Static Routing:** Routes are manually configured by network administrators.
 - **How it Works:** A fixed path is set for data to travel between networks.
 - **Pros:**
 - Simple and predictable.
 - No processing overhead on routers.
 - **Cons:**
 - Does not adapt to network failures.
 - Difficult to manage in large networks.
 - **Example:**
 - A small office network where all traffic is routed through a single gateway

Types of Routing

2. Dynamic Routing: Routes are automatically adjusted based on network conditions.

- **How it Works:** Uses **routing protocols** to exchange information and **update routes dynamically**.
- **Pros:**
 - Adapts to network changes (e.g., link failures).
 - Scalable for large networks.
- **Cons:**
 - Consumes processing power and bandwidth.
 - More complex than static routing.

Types of Routing

Common Dynamic Routing Protocols

Hop count refers to the number of intermediate routers a packet must pass through to reach its destination.

- **Distance Vector Routing** (*Based on hop count, periodic updates*)
 - **RIP (Routing Information Protocol)** → Uses hop count as a metric, updates every 30 seconds, suitable for small networks.
- **Link-State Routing** (*Maintains a map of the network, faster convergence*)
 - **OSPF (Open Shortest Path First)** → Uses Dijkstra's algorithm, supports large enterprise networks.
 - **IS-IS (Intermediate System to Intermediate System)** → Similar to OSPF, used in ISP and large networks.
- **Path-Vector Routing** (*Best for large-scale networks like the Internet*)
 - **BGP (Border Gateway Protocol)** → The backbone of the Internet, determines the best path between autonomous systems (AS).

Types of Routing

3. Default Routing: A router is configured to **forward packets to a single default route** when no specific route is found.

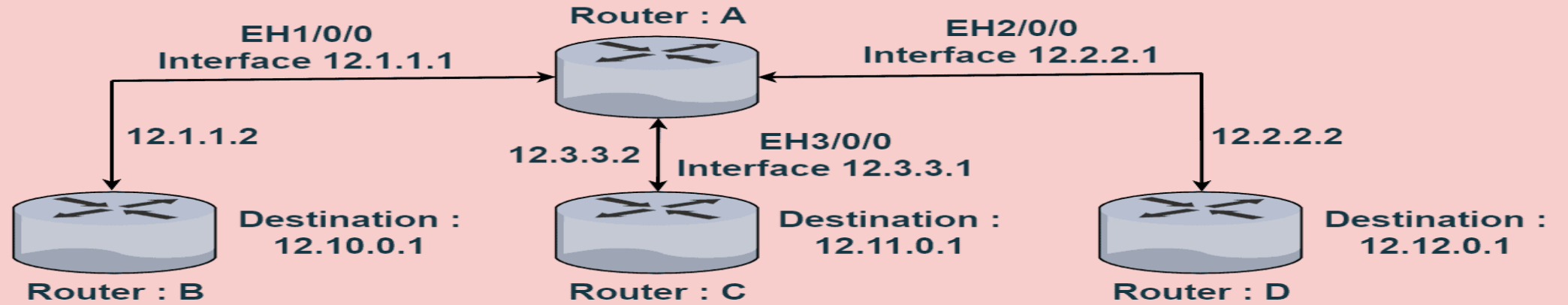
- **How it Works:** Used when a router has only one path to an external network.
- **Pros:**
 - Reduces routing table size.
 - Useful for edge devices connecting to a larger network.
- **Cons:**
 - Can lead to inefficient routing if not properly configured.
- **Example:**
 - A home router forwarding all non-local traffic to the ISP.

Routing Table

- A **routing table** is a data structure stored in a router that contains **information about available routes** for forwarding packets. It helps routers determine the best path for data transmission. It contains entries that **map destination IP addresses** to their corresponding **next-hop addresses** and metrics, such as cost or distance.
- When a router receives a data packet, it consults the routing table to determine the most **efficient path** for forwarding the packet to its destination, ensuring **optimal network performance and connectivity**.

Network Layer

Routing Table



Destination	NextHop	Interface
12.10.0.1	12.1.1.2	EH1/0/0
12.11.0.1	12.2.2.2	EH2/0/0
12.12.0.1	12.3.3.2	EH3/0/0

Routing Table

Types of Routes in a Routing Table

- **Directly Connected Routes**

- Routes to networks that are directly attached to the router.
- **Example:** If a router has an interface with IP 192.168.1.1/24, the routing table includes an entry for 192.168.1.0/24.

- **Static Routes**

- Manually configured by network administrators.

- **Dynamic Routes**

- Learned from routing protocols (RIP, OSPF, BGP).

- **Default Route (Gateway of Last Resort)**

- Used when no specific route matches a destination IP

Network Layer

How a Router Uses the Routing Table

- Receives a packet and checks its **destination IP**.
- Searches the routing table for a **matching network** entry.
- If a match is found, forwards the packet to the corresponding **next-hop router**.
- If no match is found, forwards the packet using the **default route** (if configured).
- If no valid route exists, the packet is **discarded** (ICMP "Destination Unreachable" message may be sent).
- .

Transport Layer



Introduction

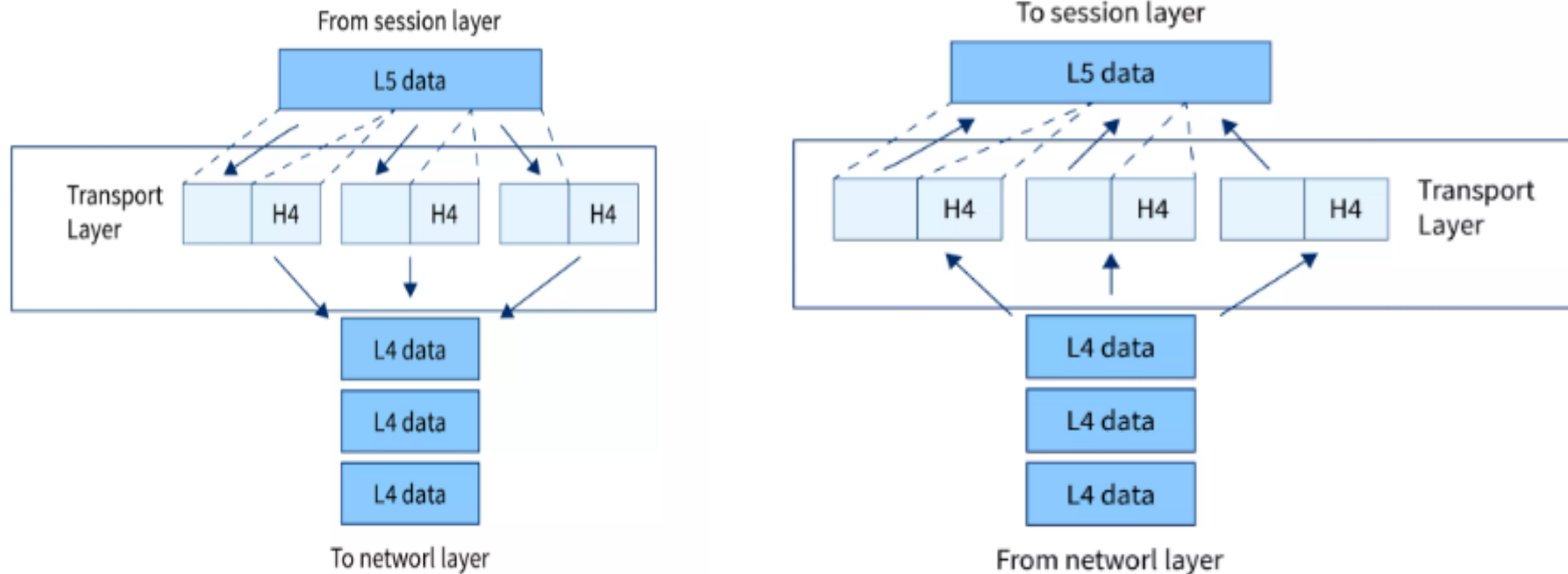
- The transport layer is the **fourth layer** of the OSI model which is responsible for the **process to process delivery of data**.
- The main aim of the transport layer is to **maintain the order** so that the data must be received in the **same sequence as** it was sent by the **sender**.
- The transport layer provides two types of services namely - **connection-oriented and connectionless**.

Transport Layer - Functions

- **Segmentation & Reassembly:** Breaks large messages into smaller segments and reassembles them at the receiver.
- **End-to-End Communication:** Ensures **reliable communication** between the source and destination devices.
- **Flow Control:** Manages **data speed** between sender and receiver to avoid overwhelming either side.
- **Error Control:** Detects and **retransmits lost or corrupted** segments for reliable delivery.
- **Connection Control:** Connection-oriented (**TCP**) and Connectionless (**UDP**):
- **Multiplexing & Demultiplexing:** Allows **multiple applications** to share a **single network** connection using port numbers.

Transport Layer

Transport Layer - Data Transmission and Working



- Data in the Transport Layer is called **Segments**.
- Transport layer is operated by the Operating System. It is a part of the OS and communicates with the Application Layer by making system calls.
- The transport layer is called as **Heart of the OSI** model.
- **Device or Protocol Use** : TCP, UDP NetBIOS, PPTP

Congestion Control

- Congestion control is a technique used to prevent **too much data** from being **sent into the network**, which can slow it down or cause packet loss.
 - It **controls traffic flow** to avoid overloading network routers and links.
 - Helps maintain **efficient network performance**.
 - Used by **protocols like TCP**, which adjust the sending rate based on **feedback** (e.g., packet loss, delay).
- **Example** : *Think of it like traffic signals in a city — regulating flow to prevent traffic jams.*

TCP – Transmission Control Protocol

- TCP (Transmission Control Protocol) is one of the **main protocols of the TCP/IP suite**.
- It lies between the Application and Network Layers which are used in providing **reliable delivery services**.
- Transmission Control Protocol (TCP) is a **connection-oriented protocol for** communications that helps in the exchange of messages between different devices over a network.
- The **Internet Protocol (IP)**, which establishes the technique for **sending data packets between computers**, works with TCP.

Features of TCP

- **Segment Numbering System:** TCP keeps **track of the segments** being transmitted or received by assigning numbers to **each and every single one of them**.
- **Connection Oriented:** It means sender and receiver are **connected to each other** till the completion of the process. The **order of the data is maintained** i.e. order remains same before and after transmission.
- **Full Duplex:** In TCP data can be **transmitted from receiver to the sender or vice – versa** at the **same time**.
- **Flow Control and Error Control**
- **Congestion Control:** TCP takes into account the **level of congestion** in the network. Congestion level is determined by the amount of data sent by a sender.

Transport Layer

UDP – User Datagram Protocol

- UDP is a **part of the Internet Protocol suite**, referred to as UDP/IP suite.
- Unlike TCP, it is **an unreliable and connectionless protocol**.
- So, there is **no need to establish a connection** before data transfer.
- The UDP helps to establish **low-latency and loss-tolerating connections** over the network.
- The UDP enables **process-to-process communication**.

Features of UDP

- **Connectionless:** Since it is connectionless, so packets are sent from **different paths** between sender and receiver.
- **Ordered delivery of data is not guaranteed:** UDP does not guarantee the order of the datagram. A datagram can be received in any order. The UDP protocol utilizes different **port numbers** for transmitting data to the correct destination.
- **Faster transmission:** UDP provides us a **faster service** of data transmission as there is no prior connection establishment before transmitting the data.
- **Acknowledgment mechanism: No Handshaking**
- **Segments are handled independently:** Every segment in UDP takes a **different path** to reach the destination. So, every UDP packet is handled **independent** of other UDP packets.

Difference Between TCP and UDP

Parameters	TCP	UDP
Connection Type	TCP is a connection-oriented protocol.	UDP is a connectionless protocol.
Speed	TCP is comparatively slower than UDP.	UDP is faster, simpler, and more efficient than TCP.
Header Size	20 bytes	8 bytes
Ordered Delivery	Yes	No
Congestion Control	Yes	No
Data Sequencing	The Transmission Control Protocol has a function that allows data to be sequenced (TCP). This means that packets arrive at the recipient in the correct order.	Unavailable since UDP uses destination and source IP to identify the right packet.
Retransmission of Data	Retransmission of lost packets are possible.	Not possible because in UDP if the packet is dropped neither the sender nor the receiver is informed about it.

Transport Layer

Difference Between TCP and UDP

Parameters	TCP	UDP
Handshake Protocols	Handshakes Protocols like SYN, SYN-ACK, ACK are available.	No Handshake protocols as it is connectionless.
Uses	TCP is generally used in different network layer protocols like HTTPS, HTTP, SMTP, POP, FTP, etc.	UDP is used in different Video conferencing apps like Zoom, streaming services like Netflix, DNS, VoIP, etc.
Error Handling	Retransmission of lost packets.	Dropping of lost packets.
Broadcasting	Not available in TCP because in case of TCP communication is done exactly between two servers i.e. host server and single recipient server.	UDP supports Broadcasting.
Examples	File transfer, web browsing, email.	Streaming media, online gaming, VoIP.

Session Layer



Session Layer

Introduction

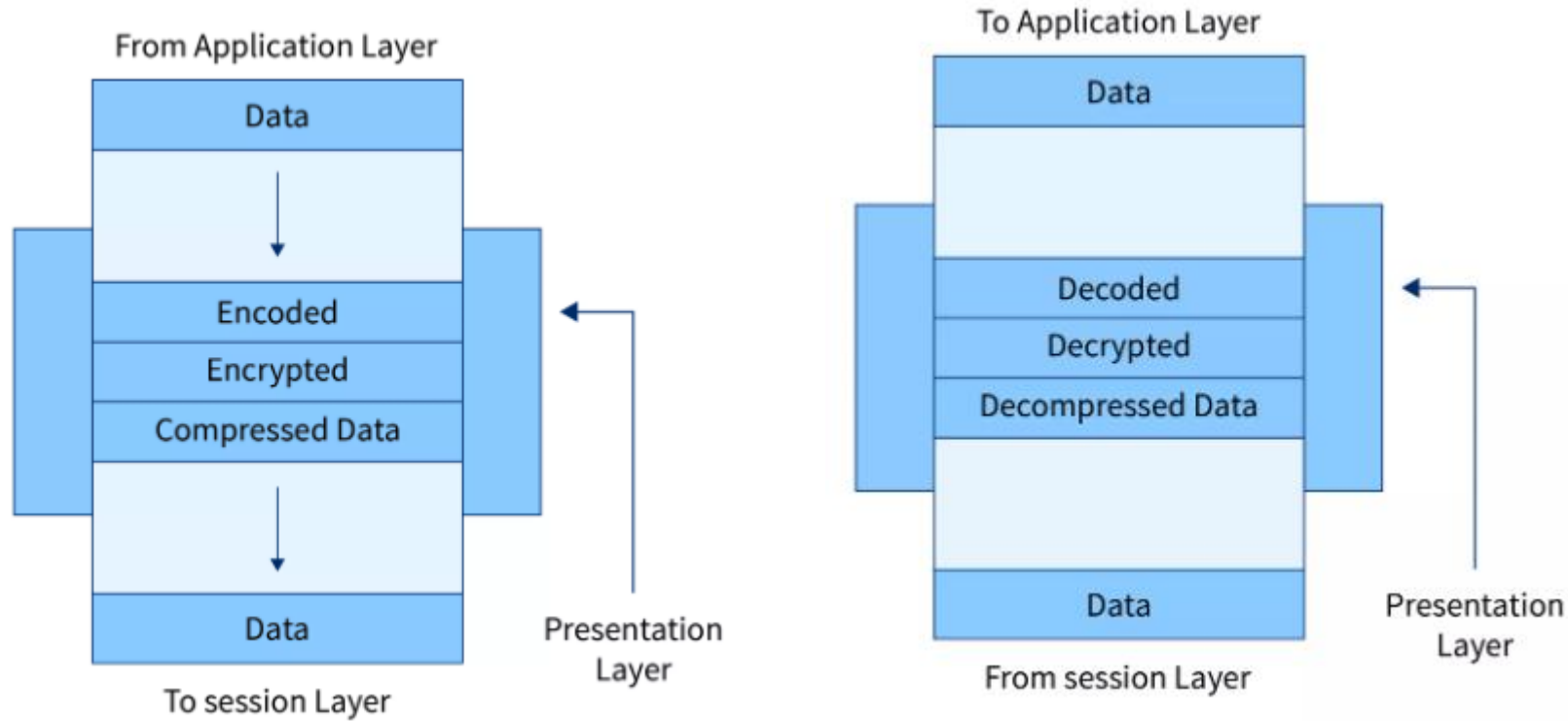
- The Session Layer can be understood as the **fifth** layer of the OSI Model.
- It is responsible for **establishing, maintaining, and ending communication sessions between applications.**
- A session is a **logical** connection between **two applications** that allows them **to exchange data** in an organized and controlled manner.

Session Layer - Functions

- **Session Management:** Handles **start, use, and end** of communication sessions.
- **Synchronization:** Adds **checkpoints** to manage errors and avoid data loss.
- **Dialog Control:** Manages communication in **half-duplex or full-duplex mode**.

Session Layer

Session Layer - Data Transmission and Working



The various protocols used in this layer are : RPC (Remote Procedure Call Protocol), RTCP (Real-time Transport Control Protocol)

Device used : Gateway

Presentation Layer



Presentation Layer

Introduction

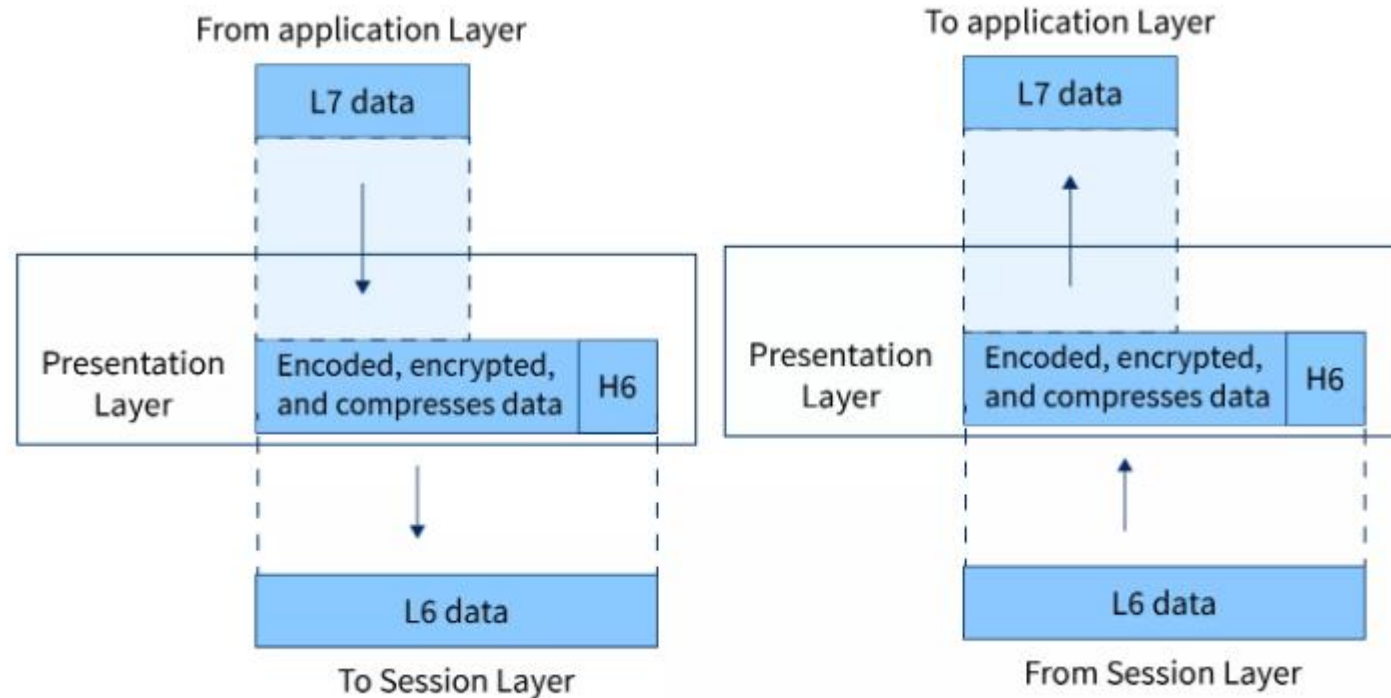
- The presentation layer is the **sixth** layer of the OSI model which mainly concentrates on the **syntax and semantics of the information** exchanged between the systems.
- The main aim of the presentation layer is to **convert** the data from **one presentation format to the other format** as different applications may use **different applications**.

Presentation Layer - Functions

- **Translation:** Converts **data formats** . **Example:** ASCII to EBCDIC.
- **Encryption/ Decryption:** Data encryption translates the **data into another form or code**. The encrypted data is known as the **ciphertext** and the decrypted data is known as **plain text**. A **key value** is used for encrypting as well as decrypting data.
- **Compression:** Reduces the **number of bits** that need to be transmitted on the network.

Presentation Layer

Presentation Layer - Data Transmission and Working



Application Layer



Introduction

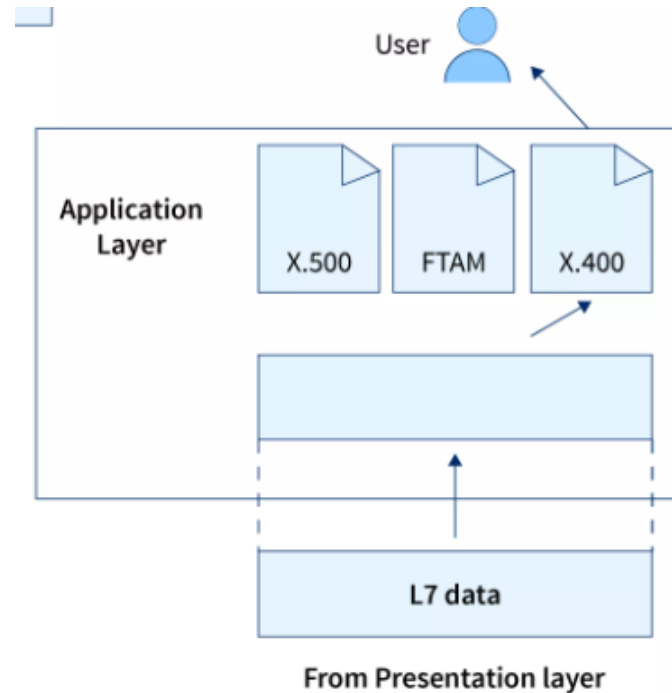
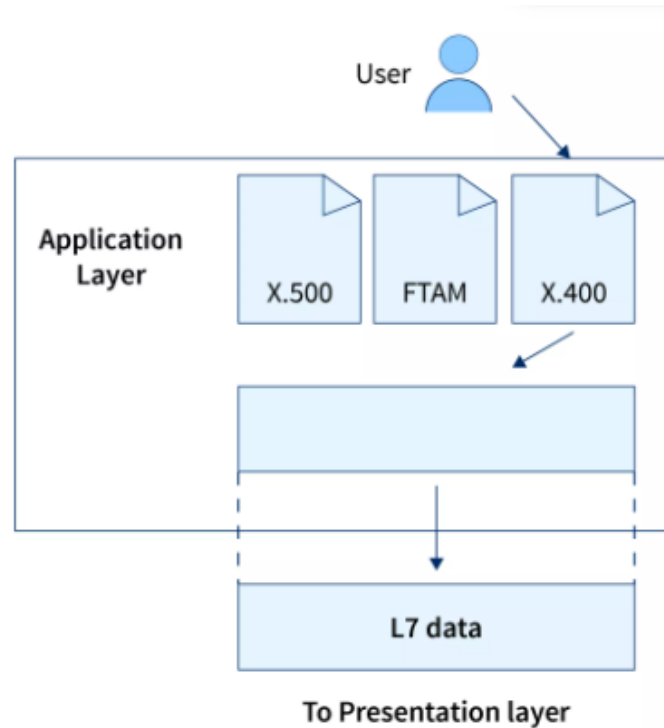
- The application layer is the **seventh** and **the last layer** of the OSI model, which mainly concentrates on **providing services to the users**.
- The application layer contains numerous protocols that are used by **users for different purposes** (refer to the list provided at the end of this section for details).
- Application layers allow users to **access and share files, access and send emails, access webpages** (via the world wide web), etc.

Application Layer - Functions

- **Network Virtual Terminal(NVT):** Allows a **user to log on to a remote host**.
- **File Transfer Access and Management(FTAM):** Allows a user to **access files** in a remote host, retrieve files in a remote host, and **manage or control files** from a remote computer.
- **Mail Services:** Provide **email service**.
- **Directory Services:** Provides **distributed database sources** and access for global information about **various objects and services**.

Application Layer

Application Layer - Data Transmission and Working



The various protocols used in this layer are : DNS (Domain Name System), SMTP (Simple Mail Transfer Protocol), FTP (File Transfer Protocol), HTTP (HyperText Transfer Protocol)

Device used : Firewalls

Application Layer

Application Layer Protocols

Protocol	Full Form	Purpose / Function	Key Point
HTTP	HyperText Transfer Protocol	Accesses and transfers web pages via client-server model	Not secure; uses port 80
HTTPS	HTTP Secure	Secure version of HTTP using SSL/TLS encryption	Encrypted; uses port 443
FTP	File Transfer Protocol	Transfers files between client and server	Not encrypted; uses ports 20 & 21
SMTP	Simple Mail Transfer Protocol	Sends emails from client to mail server	Sends only; uses port 25
IMAP	Internet Message Access Protocol	Reads/manages emails directly on server	Good for multi-device access; port 143
DNS	Domain Name System	Converts domain names to IP addresses	Internet's phonebook; uses port 53
DHCP	Dynamic Host Configuration Protocol	Automatically assigns IP addresses to devices	Reduces manual setup; uses port 67/68
SSH	Secure Shell	Secure remote login and command execution	Encrypted remote access; uses port 22

TCP/IP Model

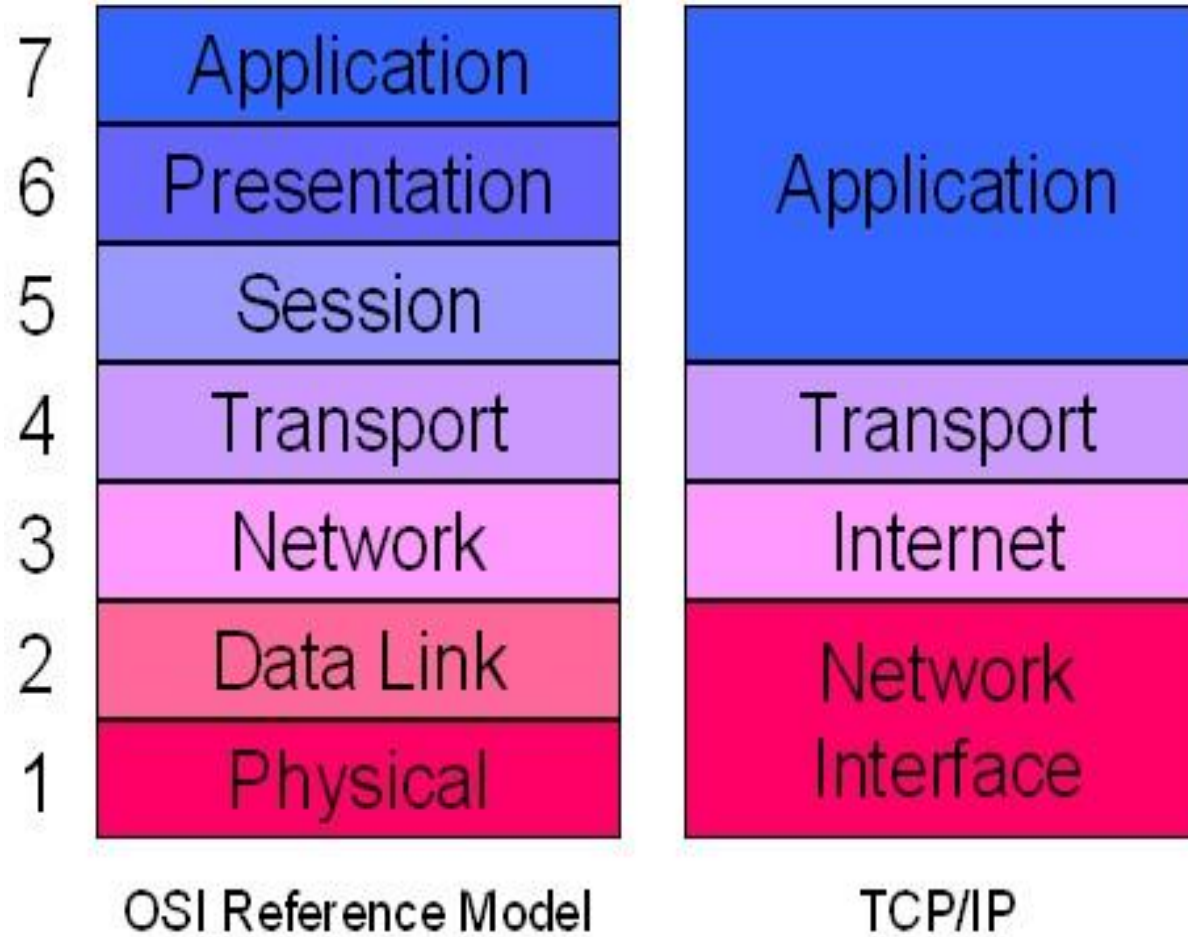


What is TCP/IP?

- TCP/IP (Transmission Control Protocol/ Internet Protocol) is the **basic communication language** or **protocol of the Internet**. Developed by **DARPA** (Defense Advanced Research Projects Agency) for the internet.
- IP defines how computers can get **data to each other over a routed**, interconnected set of networks.
- TCP defines how applications can create **reliable channels of communication** across such a network.
- Basically, IP defines **addressing** and **routing**, while **TCP** defines **how to have a conversation** across the link without garbling or losing data.

TCP/IP Model

TCP/IP Vs OSI



TCP/IP Model

TCP/IP Layers

Layer	Main Functions	Common Protocols	Responsibilities
Network Access Layer(Link Layer)	- Responsible for actual transmission of data over the physical medium; - Interfaces with hardware and defines how bits are transmitted on the network	Ethernet, Wi-Fi (802.11), PPP, ARP (also here)	- Physical (MAC) addressing - Data framing - Media access control - Error detection (CRC) - Interfacing with physical hardware
Internet Layer	Handles addressing and routing of packets across multiple networks	IP (IPv4/IPv6), ICMP, ARP, IGMP	- Logical addressing (IP addresses) - Routing decisions - Packet fragmentation/reassembly - Error reporting (ICMP) - Address resolution (ARP)

TCP/IP Model

TCP/IP Layers

Layer	Main Functions	Common Protocols	Responsibilities
Transport Layer	Ensures complete, reliable data transfer between source and destination devices	TCP, UDP	- TCP: Connection-oriented, reliable transfer - UDP: Connectionless, faster but no reliability - Port addressing - Flow & error control
Application Layer	Provides services and interfaces to user applications for network communication	HTTP, FTP, SMTP, POP3, IMAP, DNS, Telnet, SSH	- Email communication (SMTP, POP3, IMAP) - File transfer (FTP) - Web browsing (HTTP) - Domain name resolution (DNS) - Remote access (SSH)

OSI Vs TCP/IP

Feature	OSI Model	TCP/IP Model
Layers	7 Layers	4 Layers
Development	Theoretical model by ISO	Practical model by DARPA
Usage	Conceptual framework	Used in real-world networking
Flexibility	Strict layer separation	More flexible
Reliability	Defines error detection separately	Integrated into protocols like TCP
Implementation	Not widely implemented	Forms the basis of the internet

Summary:

OSI Model → Good for learning and standardization.

TCP/IP Model → Practical, widely used for networking (e.g., Internet, LAN, WAN).

In real-world networking, **TCP/IP is dominant**, while OSI is mainly used for educational and theoretical purposes.

Basics of Network Security



Network Security

- Network Security generally refers to action taken by **an enterprise or organization** to protect and secure its computer network and data. The main aim is to ensure the **confidentiality and accessibility** of the network and data.
- It safeguards the network infrastructure from **unauthorized access, misuse, modification, or denial of service attacks**.
- Involves: Protecting **hardware** (routers, switches), Securing **software** and services (applications, data transfers), Monitoring **traffic** and access, Detecting and responding to **security threats and breaches**.

Basics of Network Security

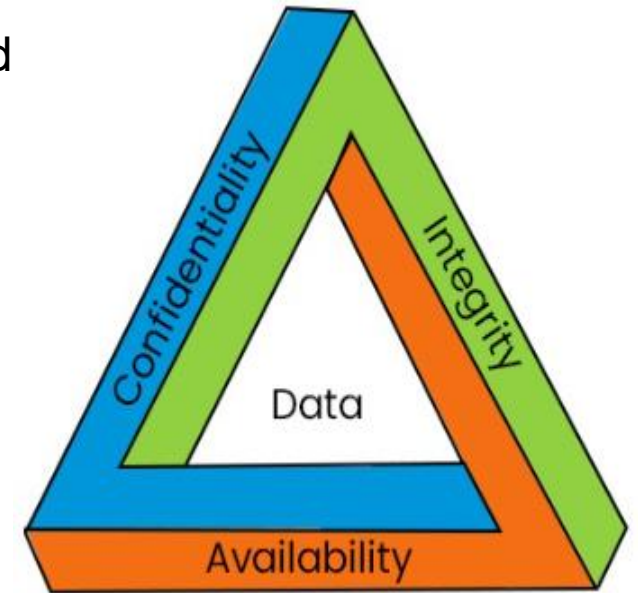
Goals of Network Security

- The CIA Triad is the foundational model for understanding and implementing effective security measures. It includes:

1. Confidentiality : Ensures that **only authorized individuals** or systems can access sensitive information. Prevents data from being **intercepted or leaked**.

Techniques used : Encryption, authentication (passwords, OTP), and access controls.

Example: Encrypting **credit card information** during online payments.



Goals of Network Security

2. **Integrity** : Ensures that the data is **not tampered with or altered** during transmission or storage.

Protects against **unauthorized modification, either malicious or accidental**.

Techniques used: Hashing, Checksums, Digital Signatures.

Example: If a file is downloaded from a website, a checksum ensures the file wasn't modified during the download.

3. **Availability** : Ensures that **systems and data** are accessible to **authorized users** whenever needed.

Protection against denial-of-service (DoS) attacks, system failures, or hardware issues.

Techniques used : Load balancing, Redundancy, Failover systems, Firewalls.

Example: Banking apps must be available 24/7 for customers to access their accounts.

Network Threats

- In network security, a **threat** is any **potential danger** that can exploit a **vulnerability** to gain **unauthorized access**, damage data, or disrupt services.
- Understanding different types of threats is crucial for **identifying risks** and applying protective measures.
- Network threats come in many forms and can target **individuals**, **companies**, or even **government systems**.
- They may originate from **hackers**, **malware**, **internal users**, or **third-party software**.

Types of Threats

1) Malware (Malicious Software) : Malware is any **software intentionally designed** to disrupt, damage, or gain unauthorized access to computer systems.

Types of Malware:

- **Viruses:** Attach themselves to files and spread when the file is run.
- **Worms:** Replicate themselves across networks without needing to attach to a file.
- **Trojans:** Disguise themselves as legitimate software but perform harmful actions in the background.
- **Ransomware:** Encrypts a user's data and demands payment (ransom) to restore access.

Example: WannaCry ransomware attack in 2017 affected thousands of systems globally.

Types of Threats

2) Phishing : Phishing involves **tricking users into revealing sensitive information** like usernames, passwords, or credit card numbers.

- Often done through **fake emails or websites** that look legitimate.
- Can also happen via **SMS (smishing) or voice calls (vishing)**.

Example: An attacker sends an email pretending to be from a bank asking the user to verify their account.

Types of Threats

3) Man-in-the-Middle (MitM) Attack : An attacker secretly **intercepts** and **possibly alters communication** between two parties without them knowing.

- Common in **public Wi-Fi scenarios** where attackers eavesdrop on data.
- Used to **steal login credentials**, **intercept** banking details, or **inject** malware.

Example: Intercepting login credentials from a user accessing their bank account over unencrypted Wi-Fi.

Types of Threats

4) Spoofing : Spoofing is when an **attacker disguises themselves as a trusted source** to gain unauthorized access or steal information.

Common Types:

- **IP Spoofing** – Fake IP address to gain network access.
- **Email Spoofing** – Forged sender address to trick users (used in phishing).
- **DNS Spoofing** – Redirect users to fake/malicious websites.
- **MAC Spoofing** – Changes MAC address to bypass controls.

Example: A hacker sends an email pretending to be your company's HR to steal login credentials.

Types of Threats

5) Denial-of-Service (DoS) & Distributed Denial-of-Service (DDoS) Attacks : These attacks aim to **overwhelm a server, service, or network** so that it becomes unavailable to users.

- **DoS**: A single system sends massive requests.
- **DDoS**: Multiple systems (botnet) are used to attack simultaneously.

Example: A DDoS attack on an e-commerce website during a sale to cause revenue loss

6) SQL Injection : Involves **inserting malicious SQL code into a database** query through user input fields. Allows attackers to bypass login, **steal or modify data, or delete databases**.

Example: Entering ' OR '1'='1 in a login field to bypass authentication.

Important Network Security Devices

Device	Function	Type	Examples
Firewall	Filters incoming and outgoing traffic based on security rules.	Hardware / Software	Cisco ASA, pfSense, Windows Firewall
Antivirus	Detects and removes malware (viruses, worms, Trojans, ransomware).	Software	Kaspersky, Norton, McAfee
IDS (Intrusion Detection System)	Detects suspicious or malicious activity in the network.	Hardware / Software	Snort, Suricata
IPS (Intrusion Prevention System)	Detects and actively blocks threats in real time.	Hardware / Software	Cisco Firepower, Palo Alto IPS
VPN (Virtual Private Network)	Creates a secure encrypted tunnel over the internet.	Software / Hardware	OpenVPN, NordVPN, Cisco AnyConnect
Proxy Server	Acts as an intermediary between users and the internet.	Software	Squid Proxy, Apache HTTP Proxy

Authentication & Access Control

- Authentication and access control are two **fundamental pillars of network security**.
- They ensure that only authorized users can **access sensitive data, systems, or services**.

1) Authentication : Process of **verifying the identity of a user, device, or system** before granting access. **Think of it as:** *"Are you really who you say you are?"*

Common Authentication Methods:

Type	Example
Something you know	Password, PIN
Something you have	OTP (One-Time Password), Smart Card, Authenticator App
Something you are	Fingerprint, Face ID, Retina Scan (Biometrics)

Authentication & Access Control

2) 2FA (Two-Factor Authentication) : A security process that requires **two types of authentication** factors for access.

- **Example:** To log in to Gmail:

1. Enter your **password** (something you know)
2. Enter **OTP** sent to your phone (something you have)

Why it matters: Adds a **second layer of defense** in case one factor (e.g., password) is compromised.

Authentication & Access Control

3) Authorization : Process of **granting permissions** to users **after authentication**, deciding what actions they can perform or what resources they can access.

Think of it as: *“You're logged in — now what can you do?”*

Key Points:

- Controls **access rights & privileges**
- Enforced using **Access Control Models**
- Common in **files, apps, systems, networks**
- **Examples:** A user can view a file but not edit it
Admin can add/delete users
Employee can access HR portal, but not finance systems

Authentication & Access Control

4) Access Control : Defines **who can access** what **within a system**, based on policies and roles.

Think of it as: *"Now that we know who you are, what are you allowed to do?"*

Model	Description	Example
RBAC (Role-Based Access Control)	Access is granted based on user roles (e.g., Admin, Guest).	Admin can delete files, Guest can only view.
DAC (Discretionary Access Control)	Owner of the resource decides who gets access.	File owner gives read permission to another user.
MAC (Mandatory Access Control)	Access based on security labels/clearance ; enforced by system policy.	Used in military or classified environments.
ABAC (Attribute-Based Access Control)	Access depends on attributes (user location, time, device).	Allow access only if user is on company Wi-Fi.

THANK YOU